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Relieving Bias in Hate Speech Detection with a Small Number of Expert Annotations: A Prompt-based Learning Approach

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Abstract

Hate speech is a major problem on social media platforms. Automatic hate speech detection methods relying on machine learning models, which learn from manually labeled datasets, have been proposed in both academia and industry. However, there is increasing evidence that hate speech detection datasets labeled by general annotators (e.g., amateurs or MTurk workers) contain systematic bias, as they cannot effectively consider language use differences among different speakers. When such biased datasets are used to train machine learning models, the resulting models will also be biased. Unlike general annotators, experts can produce much less biased annotations. However, expert annotations cannot be efficiently obtained in large quantity. This paper bridges the gap by adopting a weakly supervised learning method for hate speech detection using a small number of expert annotations. We propose a novel design that uses contrastive learning and prompt-based learning based on large language models, incorporating a group estimator, a pair generator, and knowledge injection. Using real-world Twitter posts written by African American English speakers and other racial groups as an example, extensive experiments were conducted to demonstrate the superior performance of the proposed method. The proposed approach was also evaluated on data in the LGBTQ+ community and achieved consistent results. The study has important academic and practical implications for hate speech detection and large language models.

***Disclaimer:** This article discusses and contains terms that are potentially hateful or offensive to readers. We do not censor some of these terms (though some are masked with asterisks) owing to the nature of this research.*

1 Introduction

Hate speech has become one of the major issues on social media platforms. Many methods, including advanced machine learning (ML) models (e.g., Perspective API), have been proposed for containing hate speech on social media platforms in recent years (Schmidt & Wiegand, 2017; Sap et al., 2019; Lee & Ram, 2020). Hate speech can be defined by its action, such as spreading hatred, inciting violence, or threatening people’s freedom, dignity, and safety; and by its *target*, which is a protected group (e.g., based on race, ethnicity, national origin, age, gender, sexual orientation, religion, or disability) or an individual targeted for belonging to such a group and not for their characteristics (Sanguinetti et al., 2018; Poletto et al., 2021; Cortiz & Zubiaga 2021). However, it is noted that this definition can be interpreted differently by different races, regions, and countries, in different time periods, and on different social media platforms. What is considered hateful depends on the context, such as the speaker’s social identity or dialect (Sap et al., 2019; Sap et al., 2020), the nature of the social media platforms (public vs. semi-private), and the cultural values of the country or region. Some terms or dialects, considered harmless when said by a particular group, may be perceived as hateful when being used by outsiders (Sap et al., 2019). An example is provided in Figure 1 (Röttger et al., 2021). When phrases such as “n*gga” or “b*tch” are spoken by African American English (AAE) speakers, they are commonly understood as harmless despite the fact that they should be labeled as hate speech when spoken by outsiders who are not AAE speakers (Mathew et al., 2020). Another example is the use of terms such as “queer” and “gay” which are considered harmless within the LGBTQ+ community¹ but considered offensive when used by people outside the community (Brontsema, 2004). Partly due to such *language use differences*, detecting hate speech from online writing (e.g., tweets or forum postings) is very challenging.

¹ LGBTQ+ refer to people who are lesbian, gay, bisexual, transgender, queer, and other gender identities and sexualities.



Figure 1. Annotation Results of a Tweet from an African American English (AAE) Speaker Labeled by General Annotators and Experts

In the meantime, general data annotators (e.g., amateurs or MTurk workers) often cannot discern a speaker’s social identity or dialect (Waseem, 2016). For example, it has been found that AAE texts are more likely to be falsely annotated by general annotators as hateful than the standard American English equivalents (Waseem & Hovy, 2016). Researchers refer to this phenomenon as the racial bias (Gunarathne et al., 2022) in training datasets for hate speech detection (Davidson et al., 2017; Xia et al., 2020). More generally speaking, the language use differences among different groups can lead to bias in the hate speech detection process.

Being the foundation of building ML models, reliable training datasets are crucial to the outcome of the application of ML models. Given a problematic dataset as described earlier, the bias would be amassed through the training process towards the ML models (Fu et al., 2021). Using such models, texts written by a subpopulation group may be more likely to be wrongly classified as hateful. This will result in unfairness for the subpopulation group, as a fair ML model should have a comparable performance across different subpopulation groups (Ledford, 2019; Matamoros-Fernández & Farkas, 2022; Li et al., 2023).

There is increasing attention on algorithmic bias and fairness in artificial intelligence (AI) in both computer science and information systems research (e.g., Berente et al., 2021; Teodorescu et al., 2021). Algorithms may be biased against certain groups of people and thus become unfair (Elsbach & Stigliani, 2019; Benbya et al., 2021). This is often caused by training data with biased labels, which pass the unfairness present in the data to the ML models, leading to biased applications. Consequently, organizations may make biased decisions based on these models

(Kordzadeh & Ghasemaghaei, 2021; Guynn & Wedell, 2021). To address this problem, there have been studies on how to obtain less biased datasets to build less biased ML models. For instance, annotations from experts (e.g., linguists or African American activists) have been found much less likely to wrongly label texts as hateful than general annotators (Waseem, 2016; Madukwe et al., 2020). However, expert annotations are time-consuming and expensive to obtain, and thus cannot be used efficiently (van den Broek et al., 2021; Wang et al., 2017).

Given the limited number of expert annotations, it is desirable to use ML models that do not need a large number of training data. In recent years, large language models (LLMs) have achieved great success in various text classification tasks. LLMs are transformer-based language models with a very large number of parameters (Gao et al., 2021). Such language models are often pre-trained on extensive unlabeled textual data, such as web pages, research papers, publicly available texts, and texts from books. By leveraging the knowledge gained in pre-training, LLM has the potential to learn from a small number of training data for specific tasks. This success is largely due to prompt-based learning, a weakly supervised learning approach (Liu et al., 2023). Prompt-based learning has achieved very good performance in various text classification tasks and has the potential of addressing the problem of the lack of expert-annotated training data in hate speech classification.

Therefore, a question of particular importance arises: *Can we use only a small number of data (e.g., 256 expert annotations) to train a hate speech detection model while reducing the bias caused by language use differences?* There are several methodological challenges. First, existing debiasing methods in hate speech detection often focus on removing simple racial terms. However, changing a few terms in a sentence rarely alters the dialect or language use, as these depend on sentence-level features such as linguistic patterns (Demszky et al., 2021). Therefore, even with expert annotations, how can we design a debiasing method that goes beyond term changes and addresses complex language use differences across diverse subpopulation groups?

Second, even if a debiasing method could eliminate language use differences in a sentence, there may still be a trade-off between debiasing performance and accuracy in hate speech detection. This trade-off exists because the model is trained on amateur annotations, which often contain incorrect labels (Greenstein & Zhu, 2018), potentially compromising accuracy. Using training data with expert annotations can address this issue, but such annotations are expensive to obtain. If only a small number of expert annotations are available, how can we ensure that debiasing methods consistently learn from this weak supervision with limited data—a scenario where traditional ML models struggle (Wang & Dong, 2020)?

Third, although prompt-based learning offers a feasible way to learn from weak supervision (e.g., a small number of expert annotations), it relies heavily on the *prompt* provided to the LLM and a *verbalizer* that converts the LLM output to the classification output. A good prompt can substantially impact the performance of prompt-based learning models. While some methods have been proposed in the literature, can we design a better prompt and verbalizer in order to advance the state-of-the-art performance in the context of hate speech detection? For instance, online hate speech targeting subpopulations is non-standard and ever-changing. To improve prompt-based learning in detecting hate speech, we need to inject external knowledge into the prompts to help LLMs better identify targeted groups.

To address these challenges, we propose a novel two-stage end-to-end detection framework that leverages a small number of expert annotations (e.g., 256 tweets) and addresses the language use differences with an aim to produce unbiased classification (i.e., mitigating the biases caused by language use differences). Specifically, we address the first challenge in stage one through employing contrastive learning with a novel pair generator that creates pairs of sentences differing in language use but retaining similar meanings. The last two challenges are addressed in stage two through an enhanced prompt-based learning design that incorporates external knowledge. We use AAE speakers in the U.S. on Twitter (now known as X) as a running example to illustrate the process and evaluate the performance of the proposed approach across racial

groups. Later in the paper, we also demonstrate how the method can be applied to other groups and in other contexts.

The subsequent sections of this paper are structured as follows. In Section 2, we provide a comprehensive review of biases caused by language use differences in hate speech detection and examine existing debiasing methods. Additionally, we review weakly supervised learning methods and highlight the key novelties of our study. In Section 3, we present our proposed detection framework. Section 4 outlines the experimental settings used to demonstrate the efficacy of the proposed method, and Section 5 presents the results. In Section 6, we discuss the implications and limitations of our study, and suggest directions for future research.

2 Literature Review

The issue of bias in hate speech detection arises from language use differences and limited access to expert annotations. In this section, we review the literature related to these two streams (Sections 2.1 and 2.2) and highlight the novelty of our research (Section 2.3).

2.1 Hate Speech Detection and Language Use Differences

Hate speech detection has emerged as a crucial research area, due to the increasing prevalence of online hate speech. With the recent advancements in LLMs, detection methods have significantly improved (Mozafari et al., 2020). However, studies indicate that hate speech detection algorithms often misunderstand language use differences, resulting in biases against subpopulations groups. We will examine language use differences that cause the problem of biased hate speech detection and debiasing methods in the following subsections.

2.1.1 Language Use Differences and Detection Bias

The social identity theory suggests that our self-concept is partly derived from our group memberships (Tajfel et al., 1979; Ren et al., 2012). Building upon this, the ethnolinguistic identity theory proposes that if a group perceives its language as a positive aspect of its identity, it may view other dialects or language uses, especially those of out-groups, negatively (Giles & Johnson, 1987). This can result in linguistic prejudice and discrimination, where certain speech

patterns are devalued or stigmatized based on the speaker’s group membership. For instance, as discussed earlier, words such as “n*gga” are considered hateful by AAE speakers when used by outsiders. In addition, ableist language, such as “crip” or “gimp,” can be offensive when used by non-disabled individuals but may be reclaimed positively by those with disabilities (Linton, 1998; McRuer, 2006). Similarly, in the LGBTQ+ community, terms like “queer” and “gay” have been reclaimed but can still be offensive when used by outsiders (Brontsema, 2004). Therefore, the complexity of language demonstrates how the acceptability of terms can vary by users and contexts. Understanding the influence of group membership on linguistic perception is crucial for developing information and communication technology such as hate speech detection algorithms (Lamb & Kling, 2003).

However, existing hate speech detection algorithms face challenges in understanding language use differences, which can lead to biased detection against various subpopulation groups. Specifically, this issue arises because general annotators often struggle to discern speakers’ social identities and recognize their language use differences. Consequently, they are prone to mislabeling non-hateful texts from certain subpopulations (e.g., AAE speakers or the LGBTQ+ community) as hateful. Such biases from general annotations can be passed on to hate speech detection models, resulting in biased predictions to various subpopulation groups (Xia et al., 2020). For example, studies have shown several widely used hate speech detection datasets, annotated by general workers, contain systematic racial bias, and are thus unfair to AAE speakers (Cortiz & Zubiaga, 2021; MacAvaney et al., 2019). Similarly, Oliva et al. (2021) found that the Perspective API fails to classify hate speeches correctly when words typically deemed offensive or hateful in LGBTQ+ speech are used differently or reclaimed by the community. Therefore, addressing language use differences is crucial for debiasing hate speech detection algorithms.

2.1.2 Debiasing Methods for Hate Speech Detection

Since most hate speech detection algorithms encode sentences as embeddings, existing debiasing methods focus on removing sensitive terms, such as racial terms or slurs, from the embeddings

(Xia et al., 2020; Shen et al., 2021; Cheng et al., 2021). After debiasing, hate speech detection algorithms will not encounter such information and make predictions based on genuinely hateful content (Xia et al., 2020). Early debiasing methods focus on balancing or eliminating racial terms to train a fair model. For example, Dixon et al. (2017) incorporated non-hateful data from Wikipedia to balance the distribution of racial terms. Badjatiya et al. (2019) suggested removing such terms in the text using an external knowledge base. However, such data manipulation methods cannot guarantee that the model will not retain or re-acquire such racial information during training. Therefore, researchers have explored training techniques that directly remove such information from the sentence embeddings. Among these techniques, adversarial training is a prominent method. This method operates on a dual-objective learning process, using a classifier to accurately identify hate speech while training a discriminator to make speakers' identity (i.e., language use) as undetectable as possible (Xia et al., 2020). However, the optimization process of the discriminator involves gradient reversal, which can lead to convergence issues and sensitive performance (Shen et al., 2021).

To overcome the issue in adversarial training, recent methods have incorporated contrastive learning to debias sentence embeddings (Neelakantan et al., 2022). A typical contrastive learning framework, such as SimCSE (Simple Contrastive Learning of Sentence Embeddings), trains the model on sentence pairs, which could be matched either positively or negatively depending on whether the meanings (or labels) of the two are congruent or contradictory (Gao et al., 2021). Given positive (or negative) pairs, contrastive learning will guide the encoder to minimize (or maximize) the distance between positive (or negative) pairs, leading to a better differentiation between analogous and non-analogous sentences in the embedding space. By carefully designing the training pairs, contrastive learning can remove group information from embeddings like adversarial training, and can be optimized using standard gradient descent rather than gradient reversal. Typical approaches to create such training pairs use term substitution or removal (Shen et al., 2021; Cheng et al., 2021). For instance, given a non-hateful sentence, such as "I love white."

with racial term (“white”), these methods will generate another non-hateful sentence as “I love black.” as a positive pair by replacing that racial term with an alternative value (“black”). Contrastive learning then minimizes the distance between the embeddings of positive pairs with similar semantic meanings that differ only in racial terms (e.g., “black” vs. “white”). As a result, sentences with similar semantic meanings but different racial terms will be close in the embedding space, guiding hate speech detection methods to focus on the parts of the embeddings that truly contain hateful content instead of racial terms.

However, it is important to recognize the limitations of this method. Changing a few racial terms in a sentence rarely alters the dialect or language use, as these depend on broader sentence-level features such as linguistic patterns (Demszky et al., 2021). This means that, despite term modifications, language use differences likely remain in the embeddings (Xia et al., 2020). As a result, a hate speech detection algorithm may still focus on term differences and make biased predictions based on it, even with perfect annotations (Xia et al., 2020; Shen et al. 2021).

2.2 Weakly (or Semi-) Supervised Learning

Even when language use differences are completely removed from the embedding, a hate speech detection model is still often trained on amateur annotations. Such annotations usually contain incorrect labels, which may compromise the training process of detection models, resulting in reduced accuracy. In this section, we review the literature on weakly supervised learning with incomplete supervision (Wang et al., 2020; Parnami & Lee, 2022), also known as semi-supervised learning, for hate speech detection tasks. This approach aims to learn from a small set of accurately labeled data (e.g., expert annotations) while a substantial portion of the dataset remains unlabeled or contains noisy labels (e.g., amateur annotations). This objective can be achieved by expanding the dataset through data augmentation (Hartvigsen et al., 2022) or pseudo-labeling (or denoising) (Northcutt et al., 2021; Garg et al., 2021). However, such heuristic-based methods do not guarantee sufficient label correctness for the expanded data to be used in hate speech detection tasks, potentially leading to a decrease in accuracy (Arazo et al., 2020). Given

this challenge, our attention has been directed to training classifiers using small but high-quality datasets. These techniques have improved significantly with the advancement of LLMs, particularly through fine-tuning (Sabry et al., 2022; Caselli et al. 2021; Pérez et al. 2021).

However, recent research has discovered that fine-tuning all the parameters of an LLM on small datasets (e.g., hundreds of data points) for a specific task may lead to severe overfitting issues (Brown et al., 2020; Liu et al., 2023). To overcome these issues, researchers have adopted *prompt-based learning*, where based on a given prompt, the LLMs generate a text sequence (e.g., “an extremely offensive post”) which is then mapped to a classification label of interest (e.g., “hateful”). This two-step approach reduces the reliance on tuning LLM’s parameters, thereby providing improved performance and robustness. To accommodate the two-step process, prompt-based learning methods often comprise a *prompt* and a *verbalizer* (Ding et al., 2021). Typically, the input text is first wrapped within a task-specific template, transforming the original task into a cloze-style task (Cui et al., 2022). Figure 2 provides an example of prompt-based learning employed in the context of hate speech detection tasks. A template, such as “<text> It is [MASK]”, can be utilized, where <text> denotes the placeholder for the input text that require labeling (e.g.,

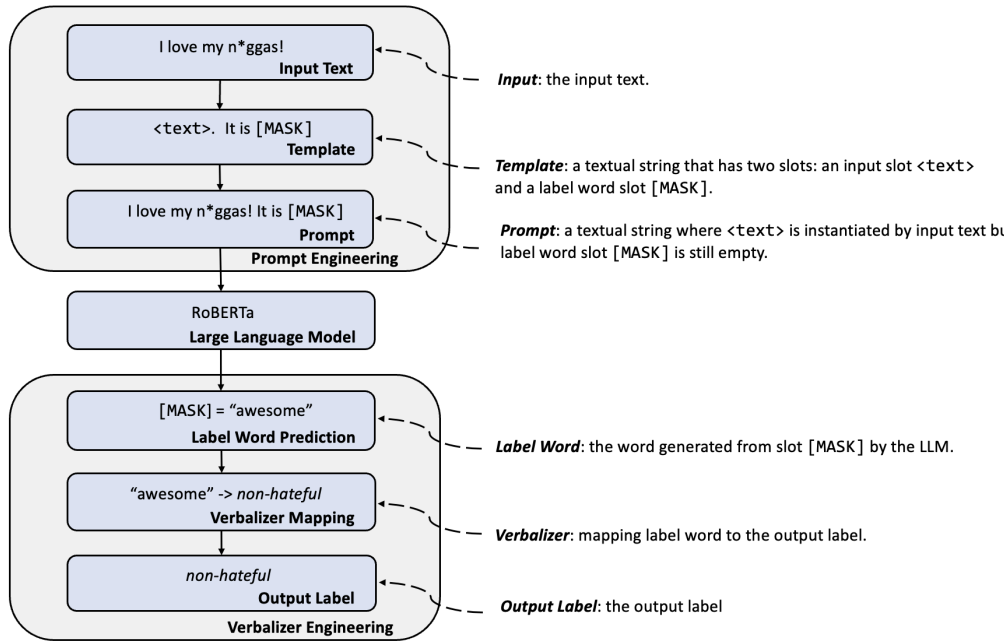


Figure 2. Example of a Prompt-based Learning Method

“I love my n*ggas!”), and [MASK] symbolizes the output text predicted by the LLM (Liu et al., 2023). The LLM then infers a word, called the label word, to fill in the [MASK] (e.g., “awesome”). Subsequently, the label word is mapped to the output label via a verbalizer (e.g., the label word “awesome” is mapped to the class *non-hateful*) for text classification. To better adapt pre-trained LLMs to specific domains, prompt-based methods can be further enhanced by injecting external knowledge. In the following subsections, we review representative methods in prompt engineering, verbalizer engineering, and external knowledge injection, respectively.

2.2.1 Prompt and Verbalizer Engineering

The objective of prompt engineering is to design a template that best wraps the input text for an LLM. Early methods rely on manually crafted templates (Petroni et al., 2020), which are expensive and time-consuming to create. Researchers have explored ways to automatically design templates, such as gradient-based search (Shin et al., 2019), prompt generation (Gao et al., 2021), and prompt paraphrasing (Haviv et al., 2021). Most prior prompt engineering methods focus on discrete templates, which are characterized in a discrete space and correspond to natural language phrases. Nevertheless, more recent studies have begun to investigate continuous (i.e., soft) prompts that directly learn the embeddings of template tokens within the model (Liu et al., 2023). For instance, Zhong et al. (2021) employ AutoPrompt, a technique which fine-tunes the initial template token embeddings with a gradient-based search to improve task accuracy. Qin & Eisner (2021) proposed automatically learning template token embeddings through a neural network, reducing need for human intervention.

After initializing a template with a sampled input text as a prompt, we can utilize an LLM to predict an intermediate label word, and then map the label word to the output label (e.g., hateful or non-hateful) through a verbalizer, as shown in Figure 2 earlier. Three major kinds of verbalizers—manual verbalizers, search-based verbalizers, and soft verbalizers—have been proposed in existing studies. Manual verbalizers use a human-defined vocabulary to link words

with output labels (Liu et al., 2023), while search verbalizers automate this process using a search algorithm (Schick et al., 2020). To reduce the effort of defining or searching in vocabularies, researchers have explored soft verbalizers. A soft verbalizer uses a continuous vector, instead of a single value, for each output label, and learn the relationship between the LLMs output vector and the label vector. For instance, Hambardzumyan et al. (2021) introduced a soft verbalizer that employs a neural network to learn label word embeddings, demonstrating advantages over traditional verbalizers. Like continuous prompts, soft verbalizers can map output from LLMs without human intervention (Cui et al., 2022).

2.2.2 External Knowledge Injection

An LLM’s ability in prompt-based learning could be further enhanced with external knowledge in specific domains that may not be covered in its pretraining (Ma et al., 2022; Wang et al., 2022). For instance, Li et al. (2022) enriched the prompt with aspect information from input sentences, demonstrating considerable promise in sentiment analysis tasks. Similarly, Hu et al. (2022) leveraged knowledge graphs to expand the vocabulary in the verbalizer for text classification. In hate speech detection, target information has been identified as important external knowledge (ElSherief et al., 2018; Silva et al., 2016). This is because the language used for hate targeting on online social media platforms is often non-standard and evolves over time, making it nearly impossible to cover all instances during training (Gröndahl et al., 2018). For example, online users may transform hate terms (e.g., “539” for “Jew” or “gei” for “gay”) to avoid censorship. Studies show that such slight modifications can significantly undermine the efficacy of detection tools like Perspective API (Gröndahl et al., 2018). Therefore, injecting external knowledge has the potential to further improve the performance of hate speech detection models based on prompt-based learning.

2.3 Key Novelty of Our Study

An overview of the literature examined above is presented in Table 1, which categorizes existing studies into debiasing methods and weakly supervised learning. The novelty of this study lies in

addressing the problem at the intersection of the two streams. As shown in the table, debiasing methods consider language use differences but struggle with small datasets, while weakly supervised learning methods excel at leveraging small datasets but may be hindered in addressing language use differences. Consequently, the extant studies are found to be ineffective for this new problem area. To address this problem, we are among the first to propose a two-stage end-to-end detection framework consisting of contrastive learning and prompt-based learning phases. Besides the detection framework, our specific original contributions include: (1) a novel pair generator for contrastive learning in the first stage, and (2) an enhanced continuous prompt with relevant external knowledge for prompt-based learning in the second stage.

Specifically, in stage one, we propose a pair generator that facilitates effective contrastive learning for our problem. For each text from a subpopulation group (e.g., AAE speakers), this method generates a Standard English pair while preserving the given sentence’s semantic meaning. By minimizing the distance between these pairs with an LLM’s embedding space, contrastive learning allows the embeddings of semantically similar sentences to be close despite language use differences. Our approach advances contrastive learning from debiasing superficial racial-related terms to sentence-level language use differences.

In stage two, given a debiased LLM, we perform hate speech detection with prompt-based learning. With a small number of data, we prefer prompt-based learning over fine-tuning as it circumvents the overfitting issues that fine-tuning may encounter. In particular, we advance the prompt design in hate speech detection by proposing an enhanced continuous prompt with external hate target information. This approach allows prompt-based learning to automatically identify the best prompt that can wrap both the input text and target information during training without human intervention. Moreover, injecting external target information into prompts further enhances the LLM’s ability to comprehend non-standard and evolving language used on online platforms, which may not be covered in its pre-training.

Methods		Addressing Language Use Differences	Learning from a Small Number of Data	Injecting External Knowledge
Debiasing	Dataset Manipulation, e.g., Badjatiya et al (2019)	No	No	Yes
	Adversarial Training, e.g., Xia et al. (2020)	Yes	No	No
	Contrastive Learning, e.g., Shen et al. (2021)	No	No	No
Weakly (or Semi-) Supervised Learning	Data Expansion, e.g., Northcutt et al. (2021)	No	Yes	No
	Fine-tuning, e.g., Gao et al. (2021)	No	Yes	No
	Prompt-based Learning, e.g., Liu et al. (2023)	No	Yes	Yes
Ours		Yes	Yes	Yes

Table 1. Comparison Between Our Method and Existing Methods

3 Methodology

Following the design science approach (Hevner et al., 2004), we develop a novel method to tackle the issue of language use differences in hate speech detection using a small number of expert

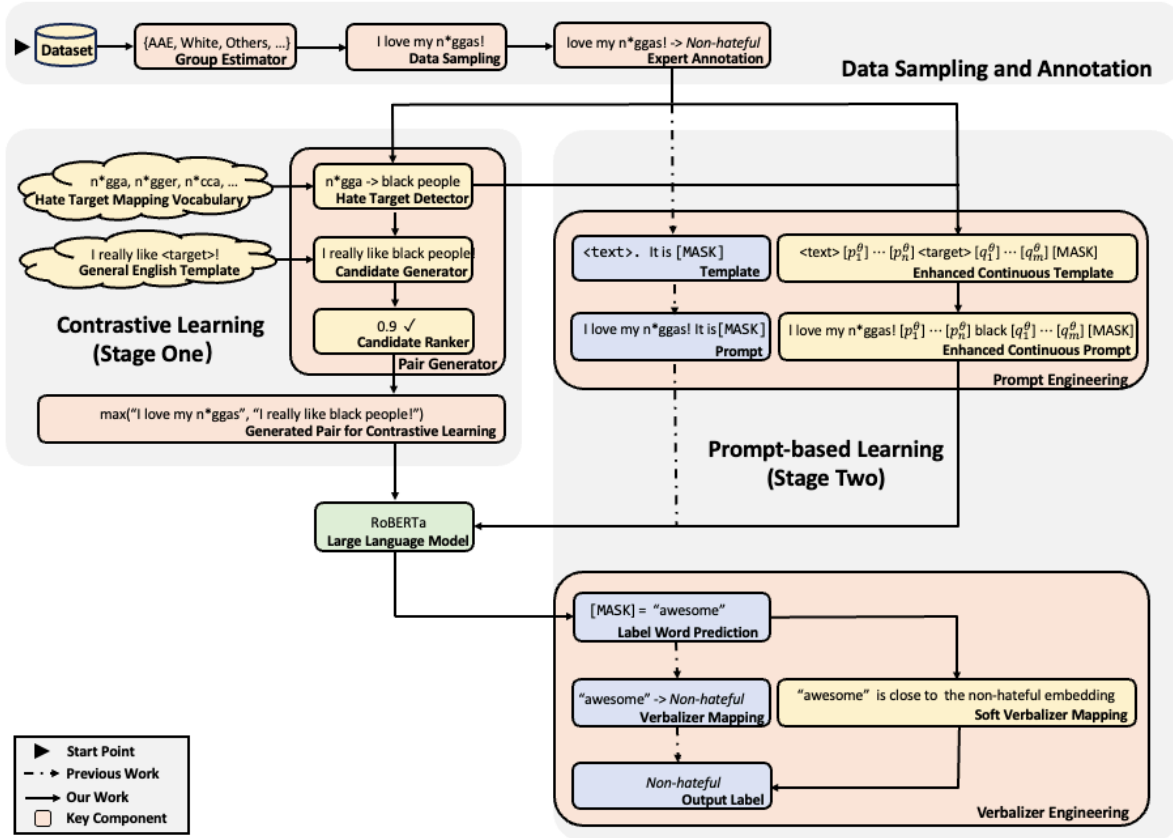


Figure 3. The Overall Workflow of the Proposed Two-stage Detection Framework

annotations. The primary goal of this method is to ensure fair performance across all groups, preventing the biased classification of relatively large amounts of non-hateful speech in any subpopulation groups, such as AAE speakers, as hateful. As discussed, we use AAE as a running example to illustrate the proposed method for simplicity, while our method is generalizable to various subpopulation groups, as outlined in the experiment sections.

The architecture overview of the proposed method is shown in Figure 3. Given a dataset, we first adopt a group estimator and propose a data sampling method to select a small number of data points (e.g., 256) that best represent the language use differences for each subpopulation group, and obtain expert annotations for these data points (Section 3.1). We then employ a two-stage end-to-end detection framework by leveraging expert annotations. In stage one, we propose a pair generator to produce a general English counterpart for each sampled text from a subpopulation group (e.g., AAE speaker). This includes a candidate generator, a hate target, and a candidate ranker. We then utilize an LLM to obtain the representations of such sentence pairs and debias it with a prevalent contrastive learning framework. This process minimizes the distance (i.e., maximizes the similarity) between these positive sentence pairs, allowing the embeddings of semantically similar sentences to be close despite differences in language use (Section 3.2). In stage two, we rely on the debiased LLM and integrate it with a prompt-based learning technique to accurately classify hate speech across different groups. The proposed prompt-based learning approach first relies on an enhanced continuous template to incorporate the input text and external hate target information from stage one as a prompt for the LLM. The LLM then predicts the output by taking the enhanced prompt and using a soft verbalizer to map the output to the final label (e.g., hateful or non-hateful). The continuous prompt and soft verbalizer are optimized simultaneously during training without human intervention (Section 3.3). In the subsequent subsections, we will present the specifics of our design.

3.1 Data Sampling and Annotation

Given a dataset, we first apply a classifier to infer the subpopulation group and its likelihood score for each text instance, referred to as the group estimator (Blodgett et al., 2016). Specifically, Blodgett et al. (2016) first defined the group categories (e.g. AAE speakers) and identified the representative geo-locations of each group (e.g., African American neighborhood blocks) using public U.S. Census data. Given the identified geolocations for each group, they then collected user profiles on Twitter from these locations and extracted their tweets as training data for the group. Given this data with group labels, they trained an ensemble model that includes a topic model. Consistent with its reported performance, extensive studies have confirmed the accuracy of racial group estimation using this estimator (Bommasani et al., 2021; Gehman et al., 2020; Weidinger et al., 2022). Once trained, the model can accurately assign a group score to each text, indicating the likelihood of the text belonging to a particular group. We then categorize the texts into their respective groups and sample a small number of those (e.g., 256) with the highest group scores. The number of samples selected from each group is determined in proportion to their representation in the dataset, ensuring a balanced and accurate reflection of the dataset’s group distribution. This procedure aids in identifying texts that most accurately represent each group. After sampling, we hire experts (e.g., linguists or African American activists) to accurately label each sample text. Given the expert annotations, we utilize them to train our two-stage end-to-end detection framework, as shown in Figure 3.

3.2 Stage One: Contrastive Learning

Following most advanced hate speech detection algorithms (Mathew et al., 2020), our proposed detection framework utilizes an LLM as the backbone model to represent input sentences as embeddings. Specifically, we chose RoBERTa, one of the most popular LLMs, to represent sentences. RoBERTa is a robustly optimized version of BERT that has gained widespread recognition for its state-of-the-art performance across various classification tasks (Liu et al., 2019). RoBERTa has been widely adopted in prompt-based learning approaches, such as LMBFF

(Gao et al., 2021), further attesting to its robustness and versatility. We first focus on fine-tuning (i.e., debiasing) an LLM to prevent it from encoding language use differences in the input sentence. To achieve this goal, we are among the first to propose generating sentence pairs with different language use while maintaining core semantics, and employing contrastive learning to fine-tune an LLM by minimizing the distance between these sentences pairs in the embedding space. For instance, a Standard English speaker might say, “I really like black people!” to express their liking to African American groups while an AAE speaker could say, “I love my n*ggas,” conveying the same semantic meaning but with different language use. Contrastive learning then minimizes the embedding distance between the AAE phrase and its Standard English counterpart, “I really like black people,” rather than semantically dissimilar Standard English phrases like “I understand black people” or “I want to support black people.” Upon completion of the training process, sentences written by people from different subgroups conveys similar meaning should be closer in the embedding space of an LLM, and vice versa. The hate speech detection could then focus on the truly hateful content in the sentence instead of differences in language use.

However, as discussed earlier, current debiasing methods with contrastive learning rely on term substitution and are unable to generate sentence pairs with different language use while retaining similar meanings, as this involves broader sentence-level features such as linguistic

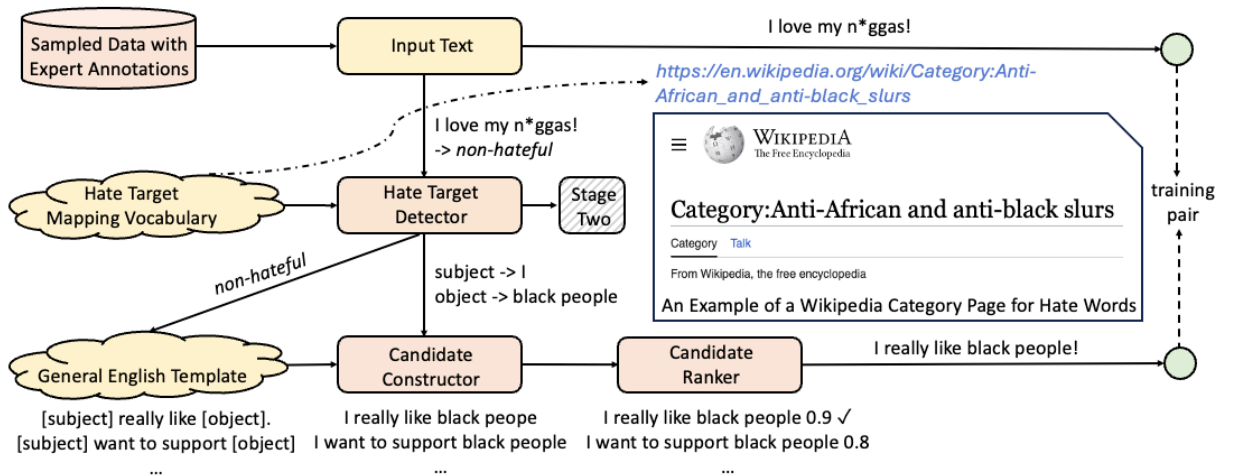


Figure 4. The Overall Workflow of the Proposed Pair Generator

patterns. To address this issue, we introduce an innovative pair generator, comprising several components such as hate target detector, candidate constructor, and candidate ranker, as depicted in Figure 4. To be more specific, given an input AAE sentence (e.g., “I love my n*ggas”) with an expert label (e.g., non-hateful), a hate target detector is first utilized to identify potential hate targets in the input AAE sentence. The hate target integrates an external hate target mapping vocabulary, which is compiled from various online sources such as Wikipedia. These detected targets are then forwarded to the candidate constructor, which generates multiple candidates with the same label (i.e., non-hateful) that likely convey a similar meaning to the input sentence. Finally, a candidate scoring and ranking module is employed to determine the most semantically similar sentence to the input, thereby constructing a positive training pair (e.g., “I really like black people!”). With the training pairs, we adopt a prevalent contrastive learning framework called SimCSE to fine-tune (i.e., debias) the LLM. The following subsections provide a detailed explanation of each component.

3.2.1 Hate Target Mapping Dictionary and Detector

The proposed pair generator first integrates a hate target detector to identify potential hate words (e.g., “n*ggas”) from the input text and map these words to the corresponding hate targets (e.g., “black”). To achieve this, we first adopt the dependency parsing technique in Stanford CoreNLP to analyze the semantic structure of the input text. We select CoreNLP as it is among the most widely-adopted dependency parsers available (Manning et al., 2014). Based on the parsing outcomes, potential hate words are identified by selecting the nominal subjects and objects within the given text. Once potential hate words have been determined, the hate target detector proceeds to match them with hate targets using a mapping vocabulary. To construct this vocabulary, we incorporate external target knowledge from various online sources including Wikipedia, Hatebase, and Urban Dictionary. We first designate several categories (e.g., black or female) as hate targets from previous work, which validates the effectiveness of these categories (ElSherief et al., 2018; Mathew et al., 2020; Shvets et al., 2021). For each hate target category, we identify

Wikipedia category pages that document its hate words. For example, as shown in Figure 4, the Wikipedia category page for Anti-African and anti-black slurs contains all sub-concepts and associated hate words. These hate words were subsequently extracted from the Wikipedia pages under each hate target category. Additionally, we supplement the vocabulary with hate words from Hatebase and Urban Dictionary to ensure it is as thorough as possible. Table 2 provides some examples of hate target and related words.

The hate target mapping vocabulary is then utilized to match hate words with their respective hate targets. In order to avoid censorship on online platforms, users often alter hate words by replacing, inserting, or removing specific characters. This includes replacement with numbers or similar pronunciation, such as “539” for “jew” and “gei” for “gay”, and character removal or insertion, like “n*g” for “n*gga” and “biitch” for “b*tch”. Simple string-matching methods are ill-equipped to handle such variations, which can hinder the effectiveness of vocabulary mapping. To address this issue, we adopt the edit distance (Ristad & Yianilos, 1998) metric in the matching process. Commonly used for automatic correction, edit distance has proven effective in detecting minor variations within words (Wang et al., 2020).

Group	Hate Target Category	Hate Words
Race	Black, White, Hispanic, Asian	n*gger; redneck; wetback; chink
LGBTQ+	Female, Lesbian, Gay, Bisexual, Transgender	b*tch; lesbian; gay; byke; transbian
Religion	Christianity, Jew, Muslim, Hindu	fundie; 539; terrorist; dothead
Disability	Physical, Mental	crip; psycho

Table 2. Examples of Hate Targets and Related Words

3.2.2 Template Mapping and Candidate Constructor

Upon identifying a hate target, we adopt a prevalent template mining technique called the MINE (Jiang et al., 2020) to extract both hateful and non-hateful templates from public sentiment datasets, such as the SST-2 (Stanford Sentiment Treebank for Binary Classification), which includes sentences with positive and negative sentiment labels. These templates, containing empty subject and object slots (e.g., “<subject> <template> <object>”), are derived from positive sentences (for hateful templates) and negative sentences (for non-hateful templates). MINE

employs a dependency parser to identify the shortest path between a sentence’s subject and object, and the phrase within this path becomes the template. A dependency parser seeks to describe the syntactic structures within a sentence by defining directed binary grammatical relations between its constituent words. This is accomplished by illustrating relations between words with labeled, directed arcs spanning from heads to dependents, displayed above the sentence. For instance, consider the sentence “I like you”. The dependency parser parses the sentence as “I *<-nsubj-like-obj->* you”. In this parsing, *nsubj* denotes the relation of subject, while *obj* indicates the object relation. From the outcome of this dependency parsing, we can generate a template that reads, “<subject> like <object>” This result enables us to understand and manipulate the sentence structure more effectively for various linguistic and computational tasks. As discussed, we use CoreNLP as our parsing tool. We then select the most frequently encountered template from positive (or negative) sentences for the proposed candidate constructor. Given an input sentence, if the sentence is labeled as hateful by experts, we propose populating the <object> slot in the hateful templates with the detected hate target through the constructor. The <subject> slot is defaulted to “I”. However, if the dependency parser identifies a distinct subject within the sentence, this subject replaces “I” in the <subject> slot.

3.2.3 Candidate Ranker and Loss Function

Once the Standard English candidates have been generated by the candidate constructor, we propose a candidate ranker to identify the best template that shares similar content or topics with the input sentence. To achieve this, we first fine-tune an LLM on Wikipedia using a next sentence prediction task. More specifically, given two sentences, we utilize the LLM to predict if the second sentence is the succeeding sentence of the first, i.e., whether they are contextually aligned. Two aligned sentences typically possess similar content or topics. After the fine-tuning process, we use the LLM to score each general candidate in relation to the input sentence. A higher score indicates that the current candidate potentially shares more similar content or topics with the

input. Finally, we select the candidate with the highest score as the output of the pair generator, which serves as the positive training pair for the input.

The above pair generator creates positive training pairs for contrastive learning, where the input sentence and the generated sentence share similar meanings. To generate negative training pairs, in which two sentences possess different meanings, we randomly select general sentences with a different label than the input sentence. For example, given a non-hateful AAE input, we randomly select a hateful Standard English sentence with a hateful label as the negative training pair. With both positive and negative training pairs, we aim to maximize the similarity (i.e., minimize the embedding distance) between the positive training pairs while minimizing the similarity (i.e., maximizing the embedding distance) between the negative training pairs (Gao et al., 2021). We adopt a prevalent contrastive learning framework called SimCSE for such a process. Specifically, given an original sentence x_i , a positive pair x_i^+ and a negative pair x_i^- , we employ the following contrastive learning loss to debias the LLM:

$$CL_i = -\log \frac{e^{\text{sim}(h_i, h_i^+)/\tau}}{\sum_{j=1}^N e^{\text{sim}(h_i, h_j^+)/\tau} + e^{\text{sim}(h_i, h_j^-)/\tau}},$$

where h_i , h_i^+ , and h_i^- represent the sentence embeddings of x_i , x_i^+ , and x_i^- obtained from the LLM, respectively. τ is a temperature hyperparameter, and the function $\text{sim}(\cdot)$ calculates the cosine similarity between the embeddings (Gao et al., 2021). N is the number of x_i in a mini-batch. The contrastive learning loss then learns to maximize the similarity between each positive pair (h_i, h_i^+) and minimize it for all other positive pairs (h_i, h_j^+) where $i \neq j$, and all negative pairs (h_i, h_j^-) , including (h_i, h_i^-) , within the mini-batch.

To the best of our knowledge, limited research has been conducted on integrating contrastive learning with prompt-based learning. We only identified a technique known as SupCon, which employs a supervised contrastive learning framework for prompt-based learning (Jian et al., 2022). SupCon incorporates various templates into the input text to generate training pairs. For instance, given an example such as “I love my n*ggas,” SupCon uses two templates: “It is [MASK]”

and “The speech is [MASK],” resulting in the training pair “I love my n*ggas. It is [MASK]” and “I love my n*ggas. The speech is [MASK].” However, without altering the input text, SupCon is unable to produce a Standard English pair for a given AAE speech. That is why we introduce a novel pair generator that can directly create Standard English pairs corresponding to AAE speech. Moreover, we are among the first to adopt the SimCSE framework, apart from SupCon, in the domain of prompt-based learning.

3.3 Stage Two: Prompt-based Learning

After debiasing the LLM with contrastive learning, we apply a prompt-based learning approach to detect hate speech. Figure 5 illustrates the overall structure of the proposed prompt-based learning technique. As mentioned, prompt-based learning comprises a prompt and a verbalizer. In terms of prompt engineering, we propose an enhanced continuous prompt approach (Hambardzumyan et al., 2021; Li & Liang, 2021; Liu et al., 2023) to automatically generate template token embeddings for hate speech detection with minimal human intervention.

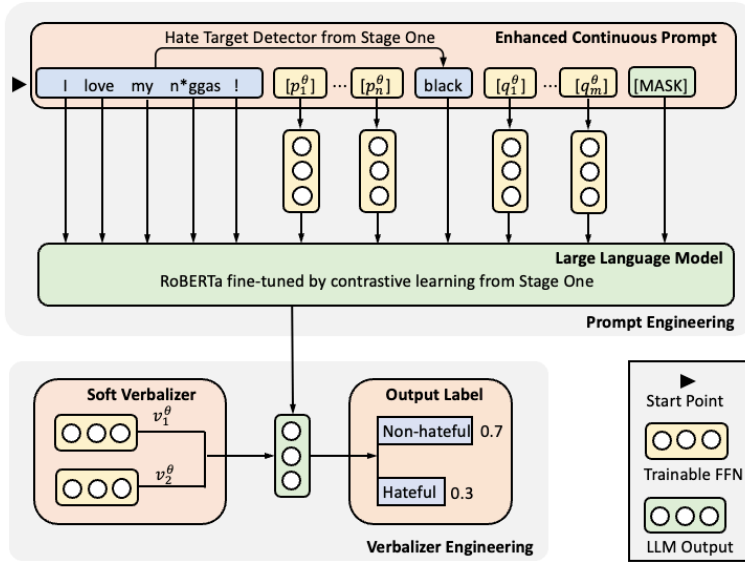


Figure 5. Overall Architecture of the Proposed Prompt-based Learning Method

Additionally, we propose incorporating hate target information from the input text to improve the enhanced continuous prompt’s performance. Regarding verbalizer engineering, we adopt a

soft verbalizer to automatically generate label word embeddings and map the LLM’s output to the final label. The following will provide a detailed explanation of each component.

3.3.1 Enhanced Continuous Prompt

A prompt-based learning method first employs a template to wrap input text into a prompt. Formally, a template is structured as follows:

$$T = \langle \text{text} \rangle \{p_1, p_2, \dots p_n\} [\text{MASK}],$$

where $\langle \text{text} \rangle$ is the slot for the input text, $[\text{MASK}]$ is the slot for the label word, and $p = \{p_1, p_2, \dots p_n\}$ is the sequence of template words which wrap $[\text{MASK}]$. For instance, let us consider the input text “I love my n*ggas.” We instantiate the template “ $\langle \text{text} \rangle$ It is $[\text{MASK}]$ ” by inserting the input text into the $\langle \text{text} \rangle$ slot, so it becomes “I love my n*ggas. It is $[\text{MASK}]$.” As discussed earlier, standard methods for selecting template words require manual intervention and expert knowledge. To minimize human involvement in prompt engineering, we implement a continuous prompt technique that automates the embedding learning for template words (Liu et al., 2023). While a standard prompt is a string formed by discrete tokens, a continuous prompt is formed by a series of embeddings of template words (i.e., no fixed words). This method begins by initializing the embeddings of the template words and subsequently fine-tunes them during training (Li & Liang, 2021; Han et al., 2022). Compared to manually defined templates, which are challenging to optimize, this method can efficiently identify optimal embeddings of template words (Hambardzumyan et al., 2021) from hate speech detection data. Formally, a continuous prompt template is structured as follows:

$$T_{original} = \langle \text{text} \rangle \{p_1^\theta, p_2^\theta, \dots, p_n^\theta\} [\text{MASK}],$$

where $\theta^p = \{p_1^\theta, p_2^\theta, \dots, p_n^\theta\}$ refers to the embeddings of template words. A single-layer feedforward neural network (FNN) is applied to learn each p_i^θ . As discussed, an effective debiased classifier should maintain accuracy for each subpopulation group without introducing unfairness towards other hate targets. Additionally, language used for hate targeting on online social media platforms is often non-standard and evolves over time, making it nearly impossible

to cover all instances during training or LLM’s pretraining. Also, online users may transform hate terms to avoid censorship. To address these concerns, we suggest enhancing the prompt by explicitly incorporating the hate target information. We begin by proposing a template that wraps the target information as follows:

$$T_{target} = \langle target \rangle \{q_1^\theta, q_2^\theta, \dots, q_m^\theta\} [MASK],$$

where $\langle target \rangle$ is the slot for hate target and $\theta^q = \{q_1^\theta, q_2^\theta, \dots, q_m^\theta\}$ is the embedding of template words that wrap the $\langle target \rangle$ slot. The hate target is identified by the hate target detector, which is part of the pair generator from the proposed contrastive learning method (Section 3.2). Given $T_{original}$ and T_{target} , we then create the enhanced template $T_{enhanced}$ by combining $T_{original}$ and T_{target} as follows:

$$\begin{aligned} T_{enhanced} &= T_{original} + target \\ &= \langle text \rangle \{p_1^\theta, p_2^\theta, \dots, p_n^\theta\} \langle target \rangle \{q_1^\theta, q_2^\theta, \dots, q_m^\theta\} [MASK]. \end{aligned}$$

More specifically, for a given input text, we first employ the hate target detector to identify the hate target. Next, we instantiate the enhanced template by populating the $\langle text \rangle$ and $\langle target \rangle$ slots with the input text and the hate target as an enhanced prompt.

3.3.2 Soft Verbalizer and Loss Function

After constructing the enhanced prompt, we utilize the LLM to predict the intermediate label word and apply a verbalizer to map it to the output label. We also adopt a soft verbalizer to find the best label word embedding for each label with minimal human involvement. Specifically, a soft verbalizer is a single-layer FNN with parameters $\theta^v = \{v_1^\theta, v_2^\theta, \dots, v_l^\theta\}$, where v_l^θ is an k -dimensional embedding vector of the label l , and L is the set of labels. In the hate speech detection task, $\theta^v = \{v_1^\theta, v_2^\theta\}$, where v_1^θ is the embedding vector of the hateful label and v_2^θ for non-hateful label.

In summary, the trained parameters in our prompt-based learning method is $\theta = \{\theta^p, \theta^q, \theta^v\}$, where θ^p and θ^q are the embeddings of template tokens that wrap the target slot $\langle target \rangle$ and

the mask slot [MASK], respectively. Furthermore, θ^v is the label word embeddings of the soft verbalizer. Following Hambardzumyan et al. (2021), given an input text x_i and its label y_i , our goal is to maximize the probability of accurately predicting y_i as follows:

$$\theta = \arg \max_{\theta} (-\log P_{\theta}(y_i|x_i)),$$

and the probability $P_{\theta}(y_i|x_i)$ is given as follows:

$$P_{\theta}(y_i|x_i) = \frac{e^{\theta_{y_i}^V \text{LLM}(T_{\theta^p,q}(x_i))}}{\sum_{l \in L} e^{\theta_l^V \text{LLM}(T_{\theta^p,q}(x_i))}},$$

where $T_{\theta^p,q}(x_i)$ is the enhanced prompt of x_i , $\text{LLM}(\cdot)$ is the output of LLM based on the enhanced prompt, and θ_l^V is the embedding of label words.

We present a comprehensive algorithm as follows:

Algorithm 1: The Proposed Method

Input: A hate speech dataset D , an LLM trained on large corpuses (e.g., Wikipedia), a set of Standard English templates T , and a mapping vocabulary of hate targets V

Output: A prompt-based learning method θ designed to detect hate speech in D

```

1  FOR each  $d_i \in D$ , DO ▷ Data Sampling and Annotation
2      Group indicator  $g_i$ , group scores  $gs_i \leftarrow \text{GROUPESTIMATOR}(d_i)$ 
3      Add  $(d_i, g_i, gs_i)$  to  $\bar{D}$ 
4  END FOR
5  Group  $\bar{D}$  into  $\{\bar{D}_1, \bar{D}_2, \dots, \bar{D}_g\}$  based on  $g_i$ , and FOR each  $\bar{D}_g$ , DO
6      Sort  $\bar{D}_g$  based on  $gs_i$  from high to low
7      Select top  $d_i \in \bar{D}_g$  to training set  $X$ 
8      Obtain expert annotations of  $X$ 
9  END FOR
10 FOR each  $x_i \in X$ , DO ▷ Contrastive Learning
11     Hate target  $h_i \leftarrow \text{HATETARGETDETECTOR}(x_i, V)$ 
12     Object  $o_i$ , subject  $s_i \leftarrow \text{DEPENDENCYPARSER}(d_i)$ 
13     Standard English candidates  $C_i \leftarrow \text{CANDIDATECONSTRUCTOR}(h_i, o_i, s_i, T)$ 
14     Training pair  $(x_i, x_i^+)$  and  $(x_i, x_i^-) \leftarrow \text{CANDIDATERANKER}(x_i, C_i, \text{LLM})$ 
15     Embeddings of training pair  $(h_i, h_i^+)$  and  $(h_i, h_i^-) \leftarrow \text{MAP}(x_i, x_i^+, x_i^-, \text{LLM})$ 
16     Update LLM by  $CL_i = -\log \frac{e^{\text{sim}(h_i, h_i^+)/\tau}}{\sum_{j=1}^N e^{\text{sim}(h_i, h_j^+)/\tau} + e^{\text{sim}(h_i, h_j^-)/\tau}}$ 
17 END FOR
18 Initialize learning parameters  $\theta = \{\theta^p, \theta^q, \theta^v\}$  ▷ Prompt-based Learning
19 FOR each  $x_i \in X$ , DO
20     Enhanced prompt  $T_{\theta^p,q}(x_i) \leftarrow x_i$ 
21     LLM  $(T_{\theta^p,q}(x_i)) \leftarrow T_{\theta^p,q}(x_i)$ 
22     Update  $\theta = \arg \max_{\theta} (-\log P_{\theta}(y_i|x_i))$ ,  $P_{\theta}(y_i|x_i) = \frac{e^{\theta_{y_i}^V \text{LLM}(T_{\theta^p,q}(x_i))}}{\sum_{l \in L} e^{\theta_l^V \text{LLM}(T_{\theta^p,q}(x_i))}}$ 
23 END FOR

```

4 Experiment Settings

We evaluate our proposed method with real-world data from Twitter, one of the largest online social media platforms. We conduct experiments to demonstrate the efficacy of our approach compared to general annotators and state-of-the-art methods. In addition to benchmarking, we conduct multiple evaluation tests focusing on individual components and other aspects of the proposed method, and provide visualizations to show its properties. Finally, we generalize the proposed method to other subpopulation groups and other contexts, such as the LGBTQ+ community. This section describes the datasets and evaluation metrics.

4.1 Datasets

We evaluate our method on the two most widely-used public hate speech detection datasets with annotations from both general annotators and domain experts. We converted all text in the two datasets to lowercase, removed control characters (such as newline and tab), and eliminated duplicate texts. Table 3 reports the detailed statistics of the processed datasets. The first dataset, WH16 (Waseem & Hovy, 2016), contains 6,799 tweets, of which 1,059 (16%) are hateful. It was initially labeled by general annotators from CrowdFlower and later re-labeled by experts, which included anti-racism activists and feminists, to obtain unbiased annotations. The second dataset, VTWK21 (Vidgen et al., 2021), is a hate speech dataset released in 2021 and contains 11,046 short text entries, of which 3,936 are hateful (36%). The data was collected from multiple sources, including Twitter and Reddit, and was annotated by experts and trained annotators. Since acquiring expert annotations is expensive and time-consuming, VTWK21 is the most prominent hate speech detection dataset reviewed by human experts to date.

Following Blodgett et al. (2016), we adopt a well-trained group estimator to identify tweets from different subpopulation groups in the datasets. For a given tweet, we use the group estimator to predict whether the tweet is written by an individual in the group AAE, Whites, or Others (i.e.,

Hispanics and Asians)². More specifically, the group estimator we adopt here is a topic model trained on over 830,000 tweets, and the reported recall for the African American group and Whites are up to 99.5% and 99.9%, respectively (Blodgett et al., 2016). Based on the estimator, we identify 1,636 (24.1%) and 2,693 (25.3%) tweets from AAE speakers in the two datasets, respectively; 4,337 (63.8%) and 7,232 (65.5%) tweets are from Whites; and the rests are from Others. The proportion of speaker groups is close to the general profile of Twitter users, with 25% of them being African Americans (Misck, 2022).

We then treat expert annotations as the gold standard and calculate the false positive rate (FPR) of general annotations for speakers in each subpopulation group (Fu et al., 2021). FPR gauges the ratio of how many non-hateful texts is falsely labeled as hateful. We find that under experts’ reviews, 22.5% and 21.3% of tweets from AAE speakers classified as hateful by general annotators are indeed non-hateful. In contrast, the other two groups are relatively well labeled. In contrast to FPR, there is no significant difference in terms of false negative rate (FNR) across the three groups. This evident difference among groups showcases a racial bias that cannot be ignored in hate speech detection.

Group	Annotation	WH16					VTWK21				
		Expert	General	Prop.	FPR	FNR	Experts	General	Prop.	FPR	FNR
AAE	Non-hateful	1341	1099	24.1%	22.5	20.3	1481	1650	25.3%	21.3	7.4
	Hateful	295	537				1312	1143			
White	Non-hateful	3767	3622	63.8%	7.5	23.9	4865	4646	65.5%	6.9	7.9
	Hateful	570	715				2367	2586			
Others	Non-hateful	632	614	12.1%	9.8	22.7	764	737	9.2%	9.7	6.8
	Hateful	194	212				257	284			

Table 3. Dataset Statistics. Prop. refers to the proportion of the dataset in percentage.

4.2 Evaluation Metrics

We report the false positive rate (FPR), false negative rate (FNR), Macro-F1 score (F1), and accuracy (ACC) to measure the performance of each method being evaluated. As stated before,

² As per the reported performance, it demonstrates promising results in distinguishing speeches from African Americans and Whites. However, Hispanics and Asians remain less discernable from each other. Consequently, we combine these two groups into one category, referred to as “Others”.

the FPR reflects the rate at which non-hateful texts are falsely classified as hateful, whereas the FNR measures how often hateful texts are falsely identified as non-hateful. Since the number of hateful and non-hateful samples in the hate speech detection datasets is heavily imbalanced, the area under the ROC curve (AUC) is not a good metric to evaluate the overall performance of the proposed method by being overly optimistic. For example, only 16% of tweets are hateful in WH16, and thus AUC will be biased towards the hateful class. Macro-F1 treats the two classes (0 for non-hateful, 1 for hateful) equally and is calculated as follows:

$$\text{Macro-F1} = \frac{1}{2} \left(\sum_{l=0}^1 \frac{2 \cdot \text{precision}_l \cdot \text{recall}_l}{\text{precision}_l + \text{recall}_l} \right),$$

where *precision* is the ratio of correctly classified hateful instances to the total classified hateful instances, and *recall* is the ratio of correctly classified hateful instances to the total actual hateful instances. We also calculate accuracy (ACC) for comparison, which is the ratio of correctly classified hateful and non-hateful instances to the total number of instances. All the metrics are reported in percentage.

Additionally, as stated before, a debiased classifier must be fair to all groups. For instance, evaluation metrics like ACC should have minimal disparity between groups such as AAE and Whites. We thus employ a commonly accepted fairness metric in machine learning, particularly in recommendation systems, known as statistical parity (SP) or demographic parity (Li et al, 2023). SP is calculated by the absolute difference in performance between two groups. A lower SP indicates better fairness. For three racial groups, we calculate the average SP (Avg SP) of ACC as:

$$\text{Avg SP} = \frac{1}{3} (|\text{ACC}_{\text{AAE}} - \text{ACC}_{\text{Whites}}| + |\text{ACC}_{\text{AAE}} - \text{ACC}_{\text{Others}}| + |\text{ACC}_{\text{Whites}} - \text{ACC}_{\text{Others}}|),$$

where ACC can be replaced with other metrics like FPR, FNR and F1.

5 Experiment Results

In this section, we report the results of our experiments. We first compare our proposed method with benchmarks in terms of the overall and fairness performance in detecting hate speech in the two datasets (Section 5.1). We then evaluate the proposed two-stage framework by examining its individual components and debiasing effectiveness (Section 5.2). We further investigate the impact of varying the size of training data and discuss computational costs alongside two distilled alternatives (Section 5.3). We also provide specific cases to demonstrate the effectiveness of our method compared to popular industry applications and chatbots on a set of commonly mislabeled data (Section 5.4). Finally, we extend the proposed method to another context, namely the LGBTQ+ community, to further assess its generalizability (Section 5.5).

5.1 Evaluation of the Performance of the Proposed Method Against Benchmarks

5.1.1 Benchmarks

We first conduct a comprehensive comparison to evaluate the overall performance of the proposed method with five sets of benchmarks. The first two sets cover common ML techniques and debiasing methods, while the last three focus on weakly supervised learning, including fine-tuning, prompt-based learning, and data expansion, as discussed in Table 1.

The first set of benchmarks focuses on common ML techniques for text classification, including TextCNN, TextRCNN, and TextRNN with attention (i.e., TextRNN-Attn). These methods serve as a control group that cannot learn from small amounts of data or address language bias. TextCNN feeds the concatenation of word embeddings into a CNN layer for classification (Gong & Ji, 2018). TextRCNN improves upon TextCNN by fusing a bidirectional RNN before the CNN layer to encode word embeddings into a sentence embedding (Lai et al., 2015). Similar to TextRCNN, TextRNN-Attn is an RNN with an attention layer that assigns greater weight to important word embeddings for improved sentence embedding (Yang et al., 2016). TextCNN, TextRCNN, and TextRNN are prevalent ML techniques that outperform traditional SVM, RNN, and CNN across various text classification tasks.

The second set of benchmarks includes debiased hate speech classifiers for addressing group information such as racial terms or language use difference, including data manipulation, adversarial training, and contrastive learning. For data manipulation, we introduce a lexical approach called Text Matching (Gitari et al., 2015), which first estimates the speaker’s group and then checks for hate words using a dictionary. When analyzing AAE speech, it excludes words targeted at the AAE group, like “n*gga,” thus mitigating language use differences. We then incorporate bias-mitigated (Dixon et al., 2017) and bias-sensitive methods (Badjatiya et al., 2019), which either remove or balance racial terms in sentences to prevent unfairly encoding these terms in the embedding. We then apply an adversarial training method to remove language information (Xia et al., 2020), the only method we identify that directly mitigates language bias in embeddings. We also include contrastive learning methods to debias LLMs, including Con (Shen et al., 2021) and FairFil (Cheng et al., 2021). They create sentence pairs based on racial terms and apply a contrastive loss to debias the LLM.

The third set of benchmarks consists of fine-tuning LLM in weakly supervised learning for hate speech detection. We first include general LLMs for text classification including BERT-based, GPT-based, and LLaMA-based LLMs. The BERT-based models include RoBERTa, the LLM used in our method. We include two RoBERTa variants with different numbers of parameters—RoBERTa-base with 123 million parameters and RoBERTa-large with 354 million parameters. In addition to BERT-based methods, we benchmark our method against prevalent architectures for LLMs, including GPT-based LLMs (Radford et al., 2019; Brown et al., 2020) and LLaMA (Touvron et al., 2023). ChatGPT, including GPT-3.5-turbo and GPT-4o, is the most advanced and widespread LLM over the past year (Chiu & Alexander, 2021). As an alternative to ChatGPT, our evaluations also include LLaMA-2-7B and LLaMA-2-13B with 7 billion and 13 billion parameters, respectively. Pre-trained on trillions of tokens, LLaMa-2 has demonstrated comparable performance to GPT-3.5 in most benchmarking.

The fourth set of benchmarks include specific LLM-based classifiers for hate speech detection, including HateXPlain (Mathew et al., 2020), HateBERT (Caselli et al., 2021), BERTweet-hate-speech (Pérez et al., 2021), Twitter-RoBERTa-Hate (Antypas & Camacho-Collados, 2023), and HaT5-Aug (Sabry et al., 2022). Studies indicate that these hate speech LLMs can significantly outperform the traditional hate speech classifiers that depend on standard deep learning techniques. Among these hate speech classifiers, HateXPlain, HateBERT, BERTweet-hate-speech, and Twitter-RoBERTa-Hate are BERT-based LLMs tailored for hate speech or tweet datasets, while HaT5-Aug is a T5-based LLM fine-tuned on a tweet hate speech dataset with data augmentation techniques.

The fifth set of benchmarks comprises the up-to-date prompt-based learning methods based on LLM in weakly supervised learning, such as PTR (Prompt Tuning with Rules), LMBFF (Better Few-shot Fine-tuning of Language Models), SupCon, and WARP (Word-level Adversarial ReProgramming). Among these, PTR simplifies prompt design by incorporating prior knowledge through logic rules, enhancing classification results in NLP tasks such as sentimental analysis (Han et al., 2022). LMBFF utilizes automatic prompt generation and pairs input instances with similar examples, leading to better learning results in text classification tasks (Gao et al., 2021). SupCon, as discussed earlier, combines prompt-based learning with contrastive learning (Jian et al., 2022). WARP, serving as the foundational architecture for our proposed prompt-based learning framework, incorporates continuous prompts (Hambardzumyan et al., 2021). It should constitute a competitive baseline to showcase the contribution of our design components other than continuous prompt-based learning.

The last set of benchmarks includes data expansion methods in weakly supervised learning, such as data augmentation and pseudo-labeling (or denoising). For data augmentation, we use the augmented hate speech dataset ToxiGen and employ chatbots including ChatGPT and HuggingChat to generate additional data based on expert annotations. For data denoising, we apply confident learning, spectral clustering, and noise modeling. Confident learning and spectral

clustering identify inconsistencies in the labeled data using statistical methods and clustering. Noise modeling adds a denoising loss for noisy data.

5.1.2 Implementation and Settings

We first discuss the implementation of our proposed method. For the backbone LLM, we chose the publicly available versions of RoBERTa-base from Hugging Face. For the hate target detector, we collected external data from Wikipedia, Hatebase, and Urban Dictionary between November 2021 and May 2022. For contrastive learning, we replicated the MINE approach from the original paper (Jiang et al., 2020) and chose CoreNLP (Manning et al., 2014) for dependency parsing. In prompt-based learning, we also implemented the enhanced continuous prompt and soft verbalizer based on the PyTorch framework and OpenPrompt toolkit (Ding et al., 2021). The number of tokens in the continuous prompt is selected by a grid search ranging from 1 to 4. The training parameters in continuous prompts and soft verbalizer are randomly initialized. We replicated all the benchmarks from the original papers. Additionally, we obtained the pre-trained weights for all LLMs used in the benchmarks from Hugging Face, except for ChatGPT. For ChatGPT, we chose the official API from OpenAI to fine-tune GPT-3.5-turbo and GPT-4o.

For dataset splitting, our method uses a random seed to divide the dataset into an 80% sampling pool and a 20% test dataset that is held out for evaluation. To evaluate the effectiveness of our approach in weakly supervised learning, we selected a small sample of data for model training. Specifically, we selected 256 samples from the pool using our sampling approach, each annotated by experts. From these sampled data, we randomly allocated 8 to a development set, leaving the remaining 248 for the training set. In training phase, we chose AdamW as the optimizer. Unlike standard ML models, the batch size in prompt-based learning should be smaller due to the limited data, so we set the batch size to 2. We set the learning rate to 0.00005 and trained the model for 5 epochs. We determined optimal parameters based on the development set’s performance. We conducted the experiments on a single Tesla V100 GPU.

The proposed method and all the benchmarks were fine-tuned using the same 256 expert annotations derived from the two datasets, WH16 and VTWK21, to ensure a fair comparison, except for data expansion methods. For data expansion, we combine the 256 expert annotations with the expanded dataset for training. Each experimental setting was repeated 20 times using random splitting seeds. Standard deviations (SD) and t-tests were employed to compare each setting with the benchmarks.

5.1.3 Main Results

We first compare our methods against the benchmarks, which encompass common ML techniques, debiasing methods, and learning approaches including fine-tuning and prompt-based learning, with data expansion excluded. All methods are fine-tuned with the same 256 expert annotations. We will discuss the data expansion benchmarks separately, as they involve additional training data.

The results are presented in Table 4. As seen in the table, the proposed method outperforms the benchmarks across all evaluation metrics ($p < 0.05$ in all cases). There are many interesting findings. First, common ML techniques, such as TextCNN, exhibit the worst performance in both datasets when compared to other benchmarks. This suggests that network-based ML techniques have limited weakly supervised learning capabilities when trained with a small number of expert annotations.

Second, debiasing methods outperform common ML techniques, showing greater potential in addressing language bias. Specifically, data manipulation methods include text matching perform worse than representation adjustment approaches like adversarial training or contrastive learning. This suggests that debiasing sentence embeddings during training is more effective than data manipulation. However, the improvement is marginal, indicating that current debiasing methods struggle when data is limited. Furthermore, we found that contrastive learning performs slightly worse than adversarial training. We contend that this is because contrastive learning relies on substituting or removing racial terms to create sentence pairs, which limits its ability to

generate pairs with different language use, as discussed earlier. Additionally, the standard errors in contrastive learning are smaller than in adversarial training, indicating more stable convergence in learning from a small number of data.

Paradigm	Method	Dataset							
		WH16				VTWK21			
		FPR	FNR	F1	ACC	FPR	FNR	F1	ACC
Common ML	TextCNN	5.5 (.25)	27.8 (1.40)	82.7 (.42)	90.0 (.56)	12.7 (.81)	12.3 (.64)	86.8 (.62)	87.8 (.49)
	TextRCNN	5.6 (.34)	28.1 (1.64)	82.0 (.59)	89.1 (.74)	12.9 (.93)	12.4 (.54)	86.5 (.45)	87.6 (.37)
	TextRNN-Attn	5.7 (.22)	28.5 (1.29)	81.4 (.51)	88.3 (.63)	13.0 (.66)	12.5 (.64)	86.4 (.54)	87.5 (.32)
Debiased Hate Speech Classifier	Text Matching	4.9 (.15)	21.5 (.66)	86.6 (.30)	90.8 (.37)	9.2 (.38)	8.8 (.32)	88.9 (.43)	90.0 (.29)
	Bias-Sensitive	5.1 (.10)	15.7 (.42)	86.4 (.35)	90.6 (.43)	9.5 (.40)	9.1 (.33)	88.7 (.46)	89.9 (.31)
	Bias-Mitigated	5.2 (.11)	16.2 (.48)	86.1 (.37)	90.2 (.46)	10.4 (.42)	9.8 (.45)	88.0 (.49)	89.2 (.37)
	Adversarial	4.7 (.09)	13.7 (.35)	87.7 (.31)	91.5 (.33)	8.6 (.36)	8.5 (.29)	89.5 (.37)	90.6 (.35)
	Con	4.8 (.06)	14.0 (.22)	87.4 (.24)	91.1 (.21)	8.7 (.20)	8.5 (.18)	89.1 (.25)	90.4 (.26)
	FairFil	4.7 (.08)	13.8 (.28)	87.6 (.20)	91.4 (.25)	8.8 (.26)	8.7 (.16)	89.0 (.30)	90.2 (.28)
Fined-tuned General LLM Classifier	RoBERTa-base	4.0 (.09)	12.5 (.33)	89.5 (.25)	92.9 (.33)	7.4 (.30)	7.2 (.15)	91.3 (.37)	92.3 (.34)
	RoBERTa-large	4.2 (.10)	13.5 (.39)	89.1 (.25)	92.4 (.25)	7.7 (.22)	7.5 (.23)	90.7 (.44)	91.6 (.18)
	GPT-3.5-turbo ³	4.0 (.10)	12.2 (.37)	90.1 (.26)	93.4 (.35)	7.2 (.18)	6.8 (.17)	92.3 (.47)	93.1 (.29)
	GPT-4o	3.9 (.11)	12.1 (.26)	90.5 (.28)	94.0 (.30)	7.0 (.29)	6.8 (.10)	92.5 (.22)	93.2 (.14)
	LLaMA-2-7B	4.0 (.09)	12.4 (.27)	89.6 (.47)	92.9 (.45)	7.2 (.13)	7.0 (.14)	92.0 (.27)	93.0 (.31)
	LLaMA-2-13B	4.1 (.09)	12.8 (.20)	89.3 (.21)	92.8 (.46)	7.3 (.17)	7.1 (.16)	92.0 (.18)	92.7 (.35)
Fine-tuned Hate Speech Classifier	HateXPlain	3.8 (.09)	12.1 (.29)	91.2 (.28)	94.7 (.20)	7.0 (.11)	6.8 (.13)	92.4 (.20)	93.1 (.25)
	HateBERT	4.2 (.09)	13.4 (.22)	89.2 (.26)	92.5 (.47)	7.7 (.14)	7.5 (.17)	90.8 (.37)	91.7 (.21)
	BERTweet	4.2 (.07)	13.5 (.31)	89.1 (.34)	92.4 (.19)	8.0 (.19)	8.0 (.15)	89.9 (.36)	91.0 (.22)
	Twitter-RoBERTa	3.8 (.07)	12.1 (.25)	90.8 (.18)	94.3 (.35)	7.2 (.11)	7.0 (.13)	92.0 (.45)	92.9 (.20)
	HaT5-Aug	4.0 (.14)	12.2 (.30)	90.5 (.31)	93.9 (.26)	7.2 (.14)	6.9 (.13)	92.1 (.28)	93.0 (.33)
Prompt-based Learning	PTR	3.9 (.05)	12.1 (.16)	90.8 (.21)	94.1 (.10)	7.3 (.12)	7.0 (.10)	92.0 (.14)	92.8 (.13)
	LMBFF	3.7 (.07)	12.1 (.25)	91.3 (.16)	94.9 (.09)	6.8 (.09)	6.7 (.08)	92.5 (.21)	93.1 (.09)
	SupCon	3.8 (.07)	12.1 (.22)	91.0 (.20)	94.4 (.16)	6.7 (.11)	6.6 (.06)	92.6 (.15)	93.2 (.08)
	WARP	3.7 (.04)	12.0 (.18)	91.3 (.11)	94.9 (.08)	6.6 (.07)	6.5 (.10)	92.6 (.12)	93.3 (.07)
General Annotators		10.1 (.25)	22.2 (.45)	79.9 (.23)	88.0 (.24)	9.9 (.18)	7.2 (.14)	90.6 (.26)	91.6 (.19)
Ours		2.5 (.03)	10.5 (.19)	93.0 (.09)	96.2 (.09)	5.4 (.07)	5.3 (.05)	93.9 (.07)	94.5 (.08)

Table 4. Evaluation of the Proposed Method and Benchmarks on the WH16 and VTWK21 datasets. Each experimental setting was repeated 20 times using random splitting seeds. The proposed method and all the benchmarks were fine-tuned using the same 256 expert annotations derived from the two datasets, WH16 and VTWK21, in each iteration. The table shows the mean results with the standard deviations in parentheses.

Third, the performance of fine-tuning general LLM-based classifier surpasses that of common ML techniques and debiasing methods in both datasets. Pre-trained on large corpora, LLM can achieve better performance when fine-tuned with a small number of annotations. We

³ GPT-3.5-turbo and GPT-4o are instruction-tuned LLMs, which is different from the other standard LLMs listed. Here, we group them together for simplicity.

also observe that the optimal performance of GPT and LLaMA architecture is closely matches that of smaller BERT-based LLMs, indicating the superior efficiency of BERT-based LLMs in classification tasks. Furthermore, within the same BERT architecture, a higher number of parameters in RoBERTa-large leads to inferior performance compared to RoBERTa-base. We postulate that this unintuitive result may arise due to the difficulty in fine-tuning larger LLMs using only a small number of expert annotations. The results validate our selection of RoBERTa-base as the backbone model over other LLM architectures such as GPT and LLaMA.

Fourth, fine-tuning advanced LLM-based hate speech classifiers yields marginal or no performance improvement compared to general LLM-based classifiers. This suggests that fine-tuning LLM on large hate speech detection datasets annotated by non-experts may lead to potential biases. Correcting such biases with a limited number of expert annotations is difficult, resulting in performance gains that remain comparable to those of general LLMs, even after fine-tuning.

Fifth, prompt-based benchmarks outperform all common ML techniques and LLMs, highlighting their exceptional capabilities in hate speech detection tasks. The performance of prompt-based learning is also consistent and does not fluctuate significantly across different implementations.

Finally, the proposed method attains the best performance across all benchmarks, including the state-of-the-art prompt-based learning method (WARP). The lowest FNR in our proposed method also suggests that the improved performance does not result from classifying more hateful and non-hateful texts, i.e., increasing the FNR. Additionally, a small standard deviation in the proposed method indicates consistent superior performance.

We then evaluate the proposed method (only using the 256 expert annotations) with the remaining data expansion benchmarks in weakly supervised learning. As discussed earlier, we compare the proposed method against data augmentation techniques, including ToxiGen, GPT-3.5-turbo, and HuggingChat, as well as data denoising techniques, including confident learning,

spectral clustering (Song et al., 2015), and noise modeling (Garg et al., 2021). It is important to note that to fully leverage the general annotations, we train these denoising methods on a mix of 256 expert annotations and 80% general annotations from our sampling pools for comparison. The results are evaluated on the 20% hold-out dataset as with the proposed method.

The evaluation results are displayed in Table 5. First, the performance of data augmentation methods still falls short of the proposed method. We contend that this is because the quality of machine-generated text is difficult to evaluate and control, thus compromising the effectiveness of training. Second, despite utilizing 256 expert annotations and 80% of the dataset as general annotations, the data denoising method underperforms compared to all data sampling and generation methods. This finding is consistent with existing studies showing that data denoising methods are not effective when the bias in annotations is dependent of data context (Garg et al., 2021), such as differences in language use between racial groups. In conclusion, using data expansion methods is more challenging than directly learning from a small number of expert annotations for bias reduction.

Paradigm	Method	Dataset							
		WH16				VTWK21			
		FPR	FNR	F1	ACC	FPR	FNR	F1	ACC
Data Augmentation	ToxiGen	4.1 (.10)	13.8 (.41)	89.6 (.25)	93.1 (.34)	7.4 (.18)	7.0 (.17)	91.9 (.37)	92.6 (.23)
	GPT-3.5-turbo	3.7 (.07)	12.0 (.47)	91.6 (.41)	94.7 (.38)	6.6 (.19)	6.6 (.26)	92.5 (.18)	93.1 (.15)
	HuggingChat	3.9 (.08)	12.6 (.33)	91.4 (.25)	93.6 (.33)	7.0 (.28)	6.8 (.14)	92.3 (.29)	92.9 (.24)
Data Denoising	Confident Learning	5.8 (.23)	23.5 (1.09)	84.7 (.55)	89.4 (.67)	10.6 (.48)	10.3 (.50)	88.0 (.48)	88.9 (.31)
	Spectral Clustering	5.7 (.24)	23.1 (.97)	84.9 (.45)	90.1 (.54)	10.6 (.40)	10.2 (.47)	88.2 (.41)	89.8 (.39)
	Noise Modelling	5.5 (.18)	22.7 (.76)	85.4 (.37)	90.2 (.41)	9.5 (.31)	9.1 (.38)	88.3 (.37)	89.8 (.21)
Ours		2.5 (.03)	10.5 (.19)	93.0 (.09)	96.2 (.09)	5.4 (.07)	5.3 (.05)	93.9 (.07)	94.5 (.08)

Table 5. Evaluation of the Proposed Method with Data Expansion Methods. Each experimental setting was repeated 20 times using random splitting seeds. The table shows the mean results with the standard deviations in parentheses.

5.1.4 Fairness Results

We conducted further evaluations on the proposed method to assess its fairness across all racial groups, and compared with the general annotations and selected benchmarks that performed the best in the main evaluation (i.e., adversarial debiasing and WARP). As discussed, we categorized the predicted results from the test dataset into three racial groups: AAE speakers, Whites, and

Others (i.e., Asians and Hispanics), employing the group estimator in Section 4.1.⁴ We then calculated the evaluation metrics for each racial group individually, and assessed the fairness across different racial groups by calculating the average statistical parity (Avg SP), where a lower value indicates greater fairness, as discussed in Section 4.2.

The results are shown in Table 6. First, we observe that the best debiased hate speech classifier (i.e., adversarial debiasing) performs the worst. Specifically, it results in notably unfair outcomes for racial groups like AAE and Others with a high Avg SP. This result is consistent with existing studies that adversarial training cannot effectively learn from a small number of data, as its bi-objective learning process requires gradient reversal, which may lead to convergence issues. As a result, adversarial debiasing lacks the ability to remove the encoding of group information from text with a bi-directional LSTM text encoder in limited data settings. In contrast, WARP and our method utilize prompt-based or contrastive learning, which can be optimized through standard gradient descent. This approach more effectively leverages a limited number of expert annotations, resulting in improved outcomes. A comparison of the proposed method with adversarial debiasing using the entire dataset will be provided later in Section 5.4.

Second, state-of-the-art methods like WARP show relatively satisfactory outcomes with a lower Avg SP, effectively leveraging a small number of expert annotations. Finally, our proposed method significantly surpasses WARP in all metrics for all groups, indicating its superior performance without sacrificing fairness for any group. Furthermore, the Avg SP in our method is half as low as that of WARP across all metrics in both datasets, indicating significant improvements in fairness across different racial groups. We attribute the lower average SP of WARP and the proposed approach to the use of continuous prompts, which help mitigate model overfitting to a specific group when only a small number of samples are available. Additionally,

⁴ It is important to note that both the proposed method and all benchmarks exclusively use the group estimator for the sampling process, and they are fine-tuned with the same sampled data.

the proposed method incorporates contrastive learning to reduce the impact of language use differences across groups, thereby lowering statistical disparities among them. In conclusion, the results show that the proposed method is indeed a debiased hate speech classifier, achieving the fairest outcomes across all racial groups.

Method	Group	WH16				VTWK21			
		FPR	FNR	F1	ACC	FPR	FNR	F1	ACC
General Annotators	AAE	22.2 (.33)	20.5 (.22)	68.0 (.30)	78.6 (.32)	20.9 (.31)	7.5 (.15)	86.9 (.33)	87.1 (.24)
	White	7.7 (.18)	24.2 (.31)	80.2 (.19)	90.0 (.21)	7.1 (.14)	7.8 (.18)	91.3 (.25)	92.1 (.16)
	Others	9.6 (.20)	22.9 (.31)	81.3 (.27)	88.3 (.24)	9.5 (.11)	6.9 (.14)	90.0 (.35)	92.8 (.23)
	Avg SP	9.68	2.44	8.87	7.59	9.20	0.56	2.93	3.79
Adversarial Debiasing	AAE	6.2 (.09)	14.0 (.39)	87.6 (.32)	89.8 (.33)	10.1 (.40)	9.1 (.32)	87.7 (.41)	89.5 (.37)
	White	3.8 (.10)	12.4 (.25)	89.1 (.26)	91.9 (.28)	7.8 (.31)	8.1 (.23)	90.1 (.33)	91.7 (.29)
	Others	4.9 (.09)	13.3 (.33)	88.4 (.28)	90.8 (.31)	8.7 (.38)	8.6 (.30)	88.8 (.35)	90.3 (.32)
	Avg SP	1.57	1.13	1.04	1.46	1.54	0.66	1.63	1.46
WARP	AAE	3.9 (.06)	11.7 (.26)	91.5 (.12)	95.0 (.10)	6.7 (.08)	6.4 (.11)	93.0 (.11)	93.3 (.08)
	White	3.6 (.04)	12.1 (.21)	91.3 (.09)	94.7 (.07)	6.1 (.06)	6.2 (.09)	92.8 (.07)	93.7 (.07)
	Others	4.2 (.05)	12.9 (.25)	90.5 (.10)	93.9 (.08)	7.1 (.09)	6.9 (.10)	92.2 (.09)	92.9 (.08)
	Avg SP	0.36	0.75	0.67	0.75	0.66	0.41	0.57	0.58
Ours	AAE	2.7 (.03)	10.2 (.21)	93.1 (.09)	96.2 (.09)	5.5 (.07)	5.2 (.04)	94.3 (.08)	94.6 (.08)
	White	2.4 (.02)	10.5 (.20)	92.9 (.10)	96.3 (.08)	5.3 (.05)	5.5 (.06)	93.9 (.07)	94.3 (.07)
	Others	2.5 (.03)	10.8 (.20)	93.2 (.10)	95.8 (.08)	5.5 (.07)	5.2 (.06)	93.8 (.08)	94.4 (.07)
	Avg SP	0.19	0.39	0.23	0.35	0.12	0.22	0.35	0.19

Table 6. Evaluation of the Proposed Method and Benchmarks on Different Racial Groups. Avg SP represents the average statistical parity; lower values mean more fairness across groups. Each experimental setting was repeated 20 times using random splitting seeds. The table shows the mean results with the standard deviations in parentheses.

5.2 Evaluation of the Proposed Two-stage Framework

5.2.1 Evaluation of Framework Components

The proposed two-stage detection framework incorporates methods include data sampling, contrastive learning and prompt-based learning. To evaluate their individual impact, we conduct experiments where each component of the proposed method was either removed or replaced with other baselines. For data sampling, we replace it with a random sampling method and a data sampling method based on embedding (Liu et al, 2023). For contrastive learning, the suggested method consists of a SimCSE framework and a pair generator. We first remove the SimCSE component and directly fine-tune the LLM with the generated training pairs. Subsequently, we replace SimCSE with the SupCon framework and the unsupervised learning technique from Gao

et al. (2021). Regarding the pair generator, we remove the candidate constructor, candidate ranker, and target detector, respectively.

For prompt-based learning, the proposed learning method comprises an enhanced continuous prompt and a soft verbalizer. Regarding the enhanced continuous prompt, we first remove the target information, reverting to the original continuous prompt method. We then replace the enhanced continuous prompt with other prominent prompt engineering techniques, including manual templates (Liu et al., 2023), mixture of soft prompt (Qin & Eisner 2021), gradient-based search (Zhong et al., 2021), prompt paraphrasing (Haviv et al., 2021), and prompt generation (Gao et al., 2021). For the soft verbalizer, we substitute it with various verbalizers, such as manual (Liu et al., 2023), knowledgeable (Hu et al., 2022), and prototypical verbalizers (Cui et al., 2022).

Method		Dataset								
		WH16				VTWK21				
		FPR	FNR	F1	ACC	FPR	FNR	F1	ACC	
Data Sampling		Random	4.2 (.09)	14.5 (.57)	88.1 (.33)	92.3 (.30)	7.9 (.19)	7.3 (.24)	91.5 (.30)	92.3 (.15)
		Embedding	3.7 (.12)	12.3 (.23)	91.5 (.28)	94.6 (.21)	6.5 (.16)	6.5 (.11)	92.5 (.31)	93.2 (.20)
Contrastive Learning	SimCSE	No SimCSE	4.3 (.15)	14.3 (.22)	88.0 (.20)	92.9 (.10)	7.5 (.18)	7.4 (.09)	91.0 (.38)	92.4 (.21)
		SupCon	4.0 (.08)	13.7 (.30)	89.0 (.43)	93.8 (.22)	7.2 (.12)	7.0 (.15)	91.4 (.34)	93.0 (.33)
		Unsupervised	3.8 (.12)	13.2 (.20)	89.9 (.31)	94.3 (.24)	7.1 (.15)	6.9 (.14)	91.5 (.18)	93.1 (.22)
	Pair Generator	No Constructor	3.7 (.12)	12.6 (.21)	90.3 (.44)	94.8 (.15)	7.0 (.13)	6.8 (.13)	91.6 (.27)	93.1 (.26)
		No Ranker	3.6 (.11)	11.7 (.22)	91.3 (.37)	95.0 (.10)	6.8 (.11)	6.6 (.13)	91.6 (.19)	93.3 (.23)
		No Detector	3.7 (.09)	12.4 (.19)	90.5 (.29)	95.0 (.21)	6.9 (.12)	6.7 (.15)	91.6 (.41)	93.2 (.12)
Prompt-based Learning	Prompt	No Target	4.1 (.09)	12.7 (.27)	91.6 (.38)	94.4 (.35)	6.5 (.12)	6.4 (.12)	92.7 (.26)	93.3 (.17)
		Manual	4.2 (.08)	12.9 (.26)	91.3 (.17)	94.3 (.25)	6.9 (.10)	6.7 (.12)	92.2 (.35)	93.0 (.10)
		Mixture	3.9 (.09)	12.5 (.30)	91.7 (.18)	94.5 (.09)	6.7 (.10)	6.5 (.13)	92.6 (.19)	93.3 (.09)
		Gradient	3.8 (.08)	12.0 (.20)	91.8 (.17)	94.8 (.38)	7.0 (.13)	6.9 (.15)	92.2 (.28)	93.0 (.11)
		Paraphrasing	3.8 (.06)	11.8 (.27)	91.9 (.25)	94.9 (.09)	6.8 (.16)	6.6 (.12)	92.5 (.27)	93.2 (.12)
		Generation	3.7 (.08)	11.7 (.19)	91.9 (.11)	94.9 (.29)	6.8 (.15)	6.6 (.15)	92.5 (.18)	93.2 (.25)
	Verbalizer	Manual	4.3 (.17)	13.1 (.24)	91.2 (.27)	94.3 (.27)	7.1 (.28)	6.9 (.22)	92.1 (.36)	92.8 (.22)
		Knowledgeable	4.1 (.13)	12.7 (.23)	91.6 (.19)	94.4 (.11)	6.8 (.17)	6.6 (.11)	92.4 (.31)	93.2 (.20)
		Prototypical	3.9 (.12)	12.2 (.28)	91.8 (.28)	94.7 (.23)	6.6 (.11)	6.4 (.13)	92.6 (.26)	93.3 (.21)
Ours		2.5 (.03)	10.5 (.19)	93.0 (.09)	96.2 (.09)	5.4 (.07)	5.3 (.05)	93.9 (.07)	94.5 (.08)	

Table 7. Evaluation of the Components of the Proposed Two-stage Framework. Each experimental setting was repeated 20 times using random splitting seeds. The table shows the mean results with the standard deviations in parentheses.

The evaluation results are shown in Table 7. For data sampling, it is observed that the performance of random or clustered sampling significantly decreases. This confirms the effectiveness of our sampling method, which accurately selects the text that best represents each

subpopulation with the highest group scores. For contrastive learning, we can observe a significant decline in performance when SimCSE is removed. Substituting SimCSE with SupCon or the unsupervised learning technique also results in a considerably lower performance compared to the proposed contrastive learning method. Removing any component of the pair generator also leads to a substantial drop in performance. Similar results were also observed for prompt-based learning. First, we observe a significant decrease in performance when removing the target information from the enhanced continuous prompt. Moreover, the performance of the enhanced continuous prompt surpasses other prompt generation methods like gradient-based search, demonstrating the effectiveness of our chosen continuous prompt approach. Second, a substantial decline is also observed when the soft verbalizer is replaced by alternative verbalizers, thus validating the superiority of the adopted soft verbalizer with minimal human intervention. In summary, the proposed two-stage framework, which incorporates SimCSE and a pair generator for contrastive learning, along with an enhanced continuous prompt and soft verbalizer for prompt-based learning, significantly outperforms other competitive baselines ($p < 0.05$ in all cases). The smallest standard deviations also underscore the robustness and stability of the proposed framework.

5.2.2 Visualization of Debiasing Effectiveness through Contrastive Learning

As discussed earlier, the proposed framework relies on contrastive learning to address the language use differences between subpopulation groups such as AAE and Standard English



Figure 6. Visualization of the Proposed Contrastive Learning on VTWK21. Standard English pairs are generated from corpus like Wikipedia.

speakers. Specifically, we use contrastive learning to maximize the similarity between texts from subpopulation groups and their corresponding Standard English counterparts. To examine the bias or disparity in LLMs and evaluate the effectiveness of debiasing, we use the VTWK21 data as an example and apply principal component analysis (PCA) to visualize the embeddings of racial groups and their Standard English equivalent, both before and after implementing contrastive learning. The results are depicted in Figure 6. From the visualization, it is evident that the embeddings of each group (blue dots for Whites, pink for AAE, and red for Others) and Standard English equivalents (cyan dots) form distinct clusters, highlighting the language use differences among different racial groups in LLMs before contrastive learning was applied. Additionally, we observe that the cluster of Whites is closest to the Standard English cluster with some overlap. In contrast, clusters of racial groups such as AAE speakers and Others are far away from the Standard English cluster with no overlap. This observation sheds light on why some LLMs favor the Whites group, and exhibit bias against other racial groups, as their embeddings primarily learn from data representing majority group. However, after applying contrastive learning, the separation between the clusters of racial groups and Standard English clusters diminishes, with their respective embeddings overlapping and no discernible boundary between them. In conclusion, our findings demonstrate that by optimizing the similarity between speech embeddings from different subpopulation groups, such as AAE speakers and their Standard English equivalents, contrastive learning effectively reduces the effect of language use differences across different groups.

5.3 Evaluation of Training Sample Sizes and Computational Costs

To evaluate the potential of our method with varying training data sizes, we first conduct a sensitivity analysis using expert annotations ranging from 10% to 80%. Specifically, we compared our method against the best debiased hate speech classifier (adversarial debiasing), and the best baseline (WARP). The learning curves for FPR and F1 on the VTWK21 dataset are presented in Figure 7. Similar trends were also observed for other performance metrics (i.e., FNR

and ACC), as well as in the WH16 dataset, but are omitted here for brevity. The results show that our method consistently outperformed both adversarial debiasing and WARP across different

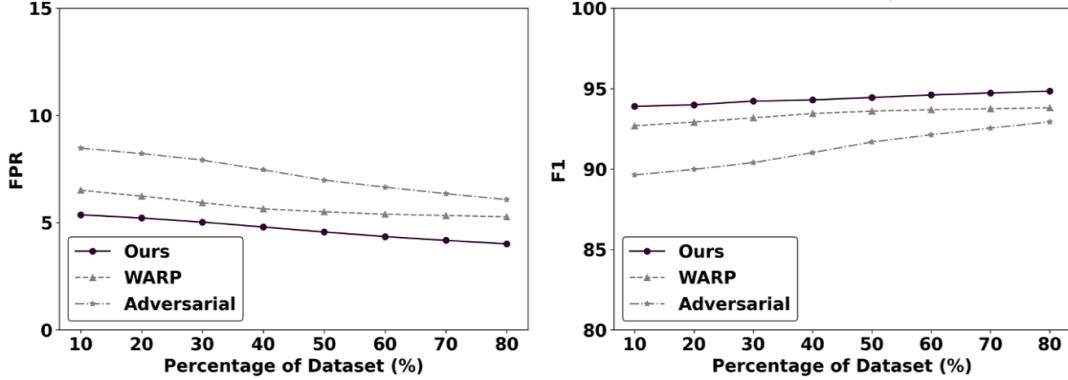


Figure 7. Learning Curves of the Training Data Size of the Proposed Method on VTWK21; “Adversarial” refers to Adversarial Debiasing training data sizes. Also, our method continued to show consistent improvement even after the percentage of data reached over 50%. This suggests that our method could benefit from utilizing more expert annotations. In summary, our approach demonstrates a distinct advantage when leveraging limited expert annotations, and shows great potential when trained with a larger number of expert annotations.

We then evaluate the computational cost of the proposed method with 256 expert annotations. As stated before, we train the proposed method using RoBERTa-base, which has 125 million parameters, as the backbone LLM on a single Tesla V100 GPU. The training time is only 30 minutes, and the inference time for one instance with up to 512 tokens takes only a fraction of a second. This shows that prompt-based learning requires only a minimal dataset size for fine-tuning while maintaining satisfactory results. To further demonstrate the effectiveness of the proposed method, we replace RoBERTa-base with distilled models as the backbone LLM, including DistilRoBERTa-base (82 million parameters) and DistilBERT-base (66 million parameters). The results are reported in Table 8. We can see the distilled LLMs require only half the training and inference time, and maintain relatively satisfactory performance in comparison to the benchmarking results in Tables 4 and 5. For instance, DistilRoBERTa-base achieves an FPR of 3.8 in WH16 and 6.8 in VTWK21, which ranks 4th and 5th out of all 30 baselines.

				Dataset
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Backbone LLM	Size (Mil)	Train. Time (Min)	Infer. Time (Sec)	WH16				VTWK21			
				FPR	FNR	F1	ACC	FPR	FNR	F1	ACC
RoBERTa	125	30	.042	2.5 (.03)	10.5 (.19)	93.0 (.09)	96.2 (.09)	5.4 (.07)	5.3 (.05)	93.9 (.07)	94.5 (.08)
DistilRoBERTa	82	16	.020	3.8 (.08)	12.1 (.28)	91.3 (.10)	94.9 (.16)	6.8 (.10)	6.3 (.06)	92.5 (.08)	93.2 (.12)
DistilBERT	66	14	.018	3.9 (.07)	12.3 (.24)	91.2 (.11)	94.7 (.14)	7.1 (.11)	6.5 (.07)	92.3 (.11)	93.0 (.10)

Table 8. Computational Analysis of the Proposed Method and Distilled Alternatives. Each experimental setting was repeated 20 times using random splitting seeds. The table shows the mean results with the standard deviations in parentheses.

5.4 Evaluation on Specific Cases

Besides the two datasets used in the previous subsections, we also conduct a study to compare the performance of the proposed method against industry applications and chatbots on a small set of cases. The dataset used, HateCheck (Röttger et al., 2021), consists of 81 tweets from AAE speakers. Despite being non-hateful in content, these tweets are frequently misclassified as hateful by existing ML models. To ensure a fair comparison, we evaluate our proposed method alongside other tools on this set of tweets, without any additional training. We select prominent industry applications for hate speech detection such as Perspective API and HateXPlain, as well as renowned chatbots or generative models like GPT-4o, GPT-3.5-turbo, BLOOM, and HuggingChat, for comparison.

The evaluation results are shown in Table 9. It is evident that, with only a limited number of expert annotations, our proposed method substantially surpasses both industry applications and widely recognized chatbots. Moreover, despite the recent advancements in chatbot capabilities, they still perform like general annotators, rather than experts, when there is no fine-tuning. Therefore, our research demonstrates the significant value of fine-tuning LLM to effectively detect hate speech like human experts.

	Perspective API	HateXPlain	BLOOM	Hugging Chat	GPT-3.5-turbo	GPT-4o	Ours
Example Cases							
This n*g’s flow too hard.			✓			✓	✓
This n*gga had so much talent, RIP.	✓				✓		✓
I’m the freshest n*gga in the room.			✓		✓	✓	✓
Me and my n*ggas ain’t playing.		✓		✓		✓	✓
All my n*ggas tryna get that money.		✓		✓		✓	✓
Why would you think it’s okay to call me a n*gger?		✓	✓				✓
Overall Performance							
ACC	80.2	84.0	86.4	86.4	86.4	87.7	93.8

FPR	19.8	16.0	13.6	13.6	13.6	12.3	6.2
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Table 9. Example Cases using HateCheck. A check (✓) means the example was correctly classified as non-hateful by the respective method. Since the examples studied here are all non-hateful tweets, F1 is the same as ACC, and FNR is not applicable.

5.5 Generalizability of the Proposed Method

To assess our method’s generalizability, we extended its application to LGBTQ+ groups, where hate speech is also common and there are language use differences in subpopulation groups. More specifically, we identify texts from a set of LGBTQ+ and non-LGBTQ+ Twitter accounts, following the literature (Oliva et al., 2021). As some accounts were banned under Twitter’s policy, we also adopt an automatic approach to identify more accounts (Karami et al., 2021). Our dataset includes 124 LGBTQ+ accounts and 187 non-LGBTQ+ accounts, with 3,600 tweets. We then trained a group estimator on a random sample of 1,800 tweets in an unsupervised manner to identify and score LGBTQ+ and non-LGBTQ+ tweets. The remaining 1,800 tweets were annotated on MTurk, each by 5 annotators, with the majority vote determining the outcomes. The tweets were then passed to experts for annotation, including LGBTQ+ members, linguists, and social science PhD students. The account profiles were also validated by experts to distinguish between LGBTQ+ and non-LGBTQ+ groups.

Method	Group	FPR	FNR	F1	ACC
General Annotators	All	14.4 (.15)	16.5 (.30)	84.6 (.31)	84.0 (.45)
	LGBTQ+	18.2 (.18)	17.6 (.25)	81.6 (.34)	80.0 (.42)
	Non-LGBTQ+	9.7 (.13)	14.4 (.32)	88.9 (.28)	87.9 (.37)
	Avg SP	8.56	3.21	7.37	7.91
Adversarial Debiasing	All	6.6 (.18)	14.7 (.39)	88.6 (.36)	89.8 (.65)
	LGBTQ+	7.1 (.20)	15.8 (.43)	87.1 (.32)	88.7 (.58)
	Non-LGBTQ+	5.9 (.11)	13.4 (.30)	89.5 (.43)	90.8 (.66)
	Avg SP	1.22	2.38	2.33	2.13
WARP	All	5.9 (.07)	12.4 (.18)	91.0 (.17)	91.7 (.15)
	LGBTQ+	6.4 (.07)	13.2 (.21)	90.2 (.18)	91.1 (.16)
	Non-LGBTQ+	5.6 (.08)	11.8 (.16)	91.6 (.14)	92.0 (.11)
	Avg SP	0.88	1.40	1.32	0.92
Ours	All	4.5 (.05)	10.4 (.15)	92.8 (.15)	93.4 (.12)
	LGBTQ+	4.5 (.05)	10.6 (.16)	92.6 (.10)	93.3 (.11)
	Non-LGBTQ+	4.5 (.06)	10.3 (.15)	92.9 (.14)	93.4 (.14)
	Avg SP	0.06	0.29	0.27	0.13

Table 10. Evaluation of the Proposed Method on the LGBTQ+ Community. Avg SP represents average statistical parity; lower values mean more fairness across groups. Each experimental setting was repeated 20 times using random splitting seeds. The table shows the mean results with the standard deviations in parentheses.

Since the dataset is smaller than WH16 and VTWK21, we choose to sample 128 data annotations for training instead of 256, using the same sampling strategy for training.⁵ We then compared the proposed method with the benchmarks using the same evaluation method as before. The results are similar to those of WH16 and VTWK21, with adversarial debiasing and WARP as two of the best baselines. The evaluation results are reported in Table 10. First, we can observe that, in comparison to expert annotations, general annotators (from MTurk) also classify more non-hateful tweets from LGBTQ+ accounts as hateful, indicating that general annotators may not consider the background of LGBTQ+ accounts. Second, the proposed method remains significantly superior to all benchmarks, highlighting the generalizability of our approach to other social groups. Third, our approach outperforms adversarial debiasing and WARP in both groups and achieves the best fairness with a significantly lower Avg SP. In summary, results from the LGBTQ+ dataset demonstrate the method’s generalizability to the LGBTQ+ community.

6 Discussions and Conclusion

6.1 Result Discussions

Hate speech detection is a challenging task because what is considered hateful depends on the speaker’s social identity and language use, while general annotators often cannot distinguish such differences. By mistakenly labeling non-hateful texts as hateful, general annotators often introduce bias into hate speech detection datasets. Meanwhile, expert annotations are less biased but they are more expensive and time-consuming to obtain. This paper proposes utilizing a small number of expert annotations to train hate speech detection models. The proposed method integrates contrastive learning and prompt-based learning, allowing the LLM to effectively detect hate speech with a small number of supervised data.

⁵ Similar results were obtained when using 256 expert annotations for training.

The proposed two-stage detection framework exhibited superior performance across all evaluation metrics, outperforming common ML techniques, debiasing methods and weakly supervised learning methods. Common ML techniques such as TextCNN displayed limitations in utilizing only a small number of expert annotations. This points to an inherent limitation of these techniques when applied to tasks that require learning from a small number of data. Debiasing methods, surpass common ML techniques in performance, a testament to their effectiveness in addressing language use differences. However, their performance is still hindered by difficulties in learning from a small number of data. Weakly supervised learning methods, including fine-tuning, prompt-based learning and data expansion, outperform all standard ML techniques and debiasing methods. This indicates their potential as a valuable tool in hate speech detection by effectively learning from a small number of data. As a recent weakly supervised learning approach, prompt-based learning outperforms traditional fine-tuning and data expansion by avoiding the need to fine-tune the entire LLM. The consistency of prompt-based learning across different implementations further highlights their robustness. However, despite the impressive performance of existing prompt-based methods, it is the proposed method that delivers the best performance across all benchmarks. The proposed method presents a balanced solution, effectively integrating the strengths of contrastive learning and established prompt-based methods including soft verbalizer and continuous prompt. It significantly mitigates language use differences between subpopulation groups with a small number of expert annotations. These findings provide empirical support for the proposed approach, which amplifies the advantages of various prompt-based learning methods. Meanwhile, this approach effectively addresses the issue of language use differences, a challenge that has been a recurrent theme in similar academic discourse.

The comparative analysis of various LLM architectures, including GPT, LLaMA, and RoBERTa, provides insightful perspectives on their debiasing capabilities and adaptability to prompt-based (or weakly supervised) learning. First, the results of the analysis underscore the

inherent challenges in fine-tuning LLMs, hinting at potential limitations of these models due to their likely dependence on large volumes of high-quality data for effective fine-tuning. This observation underscores the imperative to develop methodologies capable of harnessing limited quantities of high-quality data, such as expert annotations, to enhance the performance of LLMs. Second, this study demonstrates that smaller bi-directional LLMs, such as RoBERTa, achieves comparable performance with larger left-to-right LLMs, such as GPT and LLaMA variants. These LLMs are often used as backbone models in prompt-based learning. The comparative analysis shows the superiority of bi-directional LLMs in learning the embeddings of continuous prompts. This feature facilitates a comprehensive understanding of the input text within the prompt from both directions.

Contrastive learning also provides an innovative approach to bridge the gap between different dialects and language uses such as AAE and Standard English. This ML technique aims to learn similar representations for similar instances and different representations for dissimilar instances. The use of contrastive learning to address language use differences has shown tangible results in terms of reducing the separation between subpopulation groups. This reduction in disparity shows the importance of contrastive learning in mitigating biases present in LLMs, particularly when dealing with different dialects and language use patterns. The study reveals that contrastive learning not only aid LLMs in comprehending text in Standard English but also prevents from focusing on language uses, such as the AAE dialect. Consequently, this renders LLMs more robust and inclusive when encountering different dialectal variations and language uses. These findings align with the broader societal call for technology to be more robust and inclusive, thus facilitating greater accuracy in understanding and responding to various forms of the same language, and making our technological tools more universally effective and accessible.

6.2 Theoretical Implications

We contribute to the literature on AI fairness and algorithmic bias by proposing a novel method to reduce the bias in hate speech detection with a small number of expert annotations. In recent

years, there has been increasing attention on AI fairness and algorithmic bias in the information systems research community (e.g., Berente et al., 2021; Jain et al., 2021; Teodorescu et al., 2021). It has been suggested that the performance of AI is dependent on the quality of the input data, and the quality of such data is often questionable (Ahsen et al., 2019; Lebovitz et al., 2021). ML models trained by biased labels will inherit the unfairness from the training data, resulting in biased ML applications. This can lead to misguided decision-making in organizations and social injustice (Elsbach & Stigliani, 2019; Kordzadeh & Ghasemaghaei, 2021). Improving the fairness of AI applications is one of the key dimensions of responsible AI in information systems research (Mikalef et al., 2022), and deep learning has been a viable method to address the issue (Samtani et al., 2023). In our study, we show that the proposed method could achieve high performance by using a small number of expert annotations. Using this method, we show that the fairness of AI algorithms can be improved through the use of innovative methods to reduce bias in the training process. Our method can potentially be applied to other classification tasks.

In addition, we contribute to the literature on contrastive learning by employing this approach to address the language use differences in different subpopulation groups. To this end, we propose an innovative pair generator to create training pairs corresponding to any given sentences. In conjunction, we also utilize a template mining technique to facilitate the generation of high-quality equivalents for pairing. By optimizing the similarity between a given sentence and its Standard English counterparts, the SimCSE framework, as implemented in this study, demonstrates superior performance compared to other contrastive learning frameworks, such as SupCon.

We further enrich the literature on prompt-based learning, specifically in the field of hate speech detection. The implementation of a continuous prompt and soft verbalizer in this study requires less human involvement in comparison to other methods. The performance of our proposed method outperforms existing benchmarks, achieving state-of-the-art results. Additionally, we integrate a hate target detector and external data sources, which allow the

continuous prompt to effectively recognize various hate targets. Comparing with some proposed algorithms in other domains (e.g., peer-to-peer lending) that obtain fairness for other subpopulation groups at the cost of accuracy (Fu et al. 2021), our proposed method successfully achieves satisfactory outcomes while preserving, and even enhancing, accuracy and F1 scores as well as fairness as measured by the average statistical parity.

Our research also contributes to the literature on how digital technologies are shaping hate crimes (Chau & Xu, 2007; Chan et al., 2016). By studying the problem of hate speech detection using ML models and LLMs, we uncover how digital technologies can introduce racial bias in the process and how such bias can be mitigated. This improves our understanding of the formation of racial bias and discrimination in AI and how to address it. Our proposed methods and findings can be applied in a wide range of problems related to social justice and other issues.

6.3 Practical Implications

Our research provides several design recommendations for debiasing hate speech detection algorithms. First, our findings show that reliance on amateur annotations often leads to biased outcomes against protected groups in hate speech detection. This underscores the importance of seeking expert annotations to train fair models for subpopulations with different language use. Second, prevalent debiasing methods that focus on removing racial terms do not adequately address the language use differences among subpopulations. Our approach demonstrates that explicitly addressing such biases can effectively handle this language use difference, leading to fairer detection across different groups. Third, our results indicate that smaller LLMs like RoBERTa, when paired with prompt-based learning, can outperform larger LLMs such as ChatGPT for hate speech detection tasks. This suggests that platforms dealing with high volumes of hateful content could adopt smaller LLMs like BERT for more efficient and cost-effective hate speech detection without sacrificing performance.

Our research also suggests several policy recommendations for investigating the bias in hate speech detection algorithms, which is essential for content moderation on digital platforms (FRA,

2023; New American, 2024). First, our findings show that differences in language use are a significant source of bias in hate speech detection. Policymakers should support debiasing algorithms that address these differences rather than simply removing racial terms. Second, policymakers should mandate detailed audits to evaluate detection systems’ performance across different subpopulations using fine-grained evaluation metrics, such as statistical parity, to ensure consistent fairness across. Third, policies should require full transparency in the development of hate speech detection algorithms, including the specifics of data collection and training methods, rather than focusing solely on the final model weights.

Our method can further help data annotation companies and online data annotation platforms such as MTurk and CrowdFlower evaluate annotators’ work. These data annotation companies and platforms rely heavily on the quality of their annotators’ work to provide accurate and reliable annotations to their clients. However, due to the subjective nature of annotation, it is common for annotators to introduce biases into their work, which can result in inaccurate and unreliable data. Our method offers a solution to this problem by providing a means of evaluating annotators’ work and identifying those who consistently make biased annotations. Data annotation companies and platforms can take steps to improve the quality of their workforce by rewarding better annotators through evaluation. This, in turn, will enable the platforms to provide less biased annotations to their clients, ultimately improving the accuracy and reliability of their services.

6.4 Limitations and Future Directions

Our proposed method can be extended in several ways. First, while we currently use a convenient sampling method based on group estimators for expert annotations, future research could explore more advanced techniques, such as using amateur annotations to prioritize data samples with higher uncertainty for expert review. This approach could enhance model performance, as samples difficult for amateurs may also challenge the model. Second, although this study focuses on racial and LGBTQ+ groups, incorporating datasets representing other marginalized communities (e.g., disability, age, or intersectional identities) and multimedia (e.g., racial hatred

photos) data would provide a more comprehensive understanding of biases in social media content. It would also be interesting to investigate whether the proposed method can be extended to other languages by expanding the dataset beyond English-speaking environments, as certain words or phrases considered non-hateful in English may be interpreted as hateful in other languages. Third, to address the evolving nature of hate speech, our model could benefit from automatic methods to continuously update hate target mapping vocabularies from social media. Fourth, while we currently use contrastive learning to mitigate language biases in sentence embeddings, future work could explore other debiasing techniques, such as denoising amateur annotations or stabilizing adversarial training performance by leveraging expert annotations. These directions could help our method continuously support the development of debiasing hate speech detection in social media analysis.

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